47 JEE ADVANCED

YEARS (2024 TO 1978)

CHAPTER-WISE & TOPIC-WISE SOLVED PAPERS

FIRST TIME EVER!

Analyse the question-wise difficulty level in real-time with Correct (C), Wrong (W) and Unattempted (UA) questions response tagging provided by IIT-JEE

MATHEMATICS

ANSWER KEY VERIFIED FROM OFFICIAL WEBSITE OF JEE ADVANCED

JEE ADVANCED 2024 Paper Analysis

Paper 1

Section	Question Type	No. of Questions	Marks per Question	Negative marks per question	Total Marks (Section wise)
1	Single Correct MCQs	4	3	–1	12
2	More than One Correct MCQs	3	4	–2	12
3	Non Negative Integer Type Questions	6	4	0	24
4	Matrix Type Questions	4	3	–1	12
	Total Questions per subject	17			60

Paper 2

Section	Question Type	No. of Questions	Marks per Question	Negative marks per question	Total Marks (Section wise)
1	Single Correct MCQs	4	3	–1	12
2	More than One Correct MCQs	3	4	–2	12
3	Non Negative Integer Type Questions	6	4	0	24
4	(Numerical Value Based Questions)	4	3	–1	12
	Total Questions per subject	17			60

Total No. of Questions in Part 1 = 51 Questions

Total No. of Questions in Part 2 = 51 Questions

Each Subject Carries = 60 Marks

Total Marks = 180

Paper Analysis

Paper Difficulty Compared to Last Year	Easier Compared to 2023 Paper
Overall Difficulty Level of JEE 2024	
Easy	18%
Moderate	49%
Difficult	33%

Mathematics Topics	Paper 1 Difficulty Level	Paper 2 Difficulty Level	Remarks
Basic maths	1 moderate question from basic mathematics	no question from this topic in paper 2	
Functions & Relations	1 easy question from this topic	Total 5 topics from Functions & relations out of which, 1 is difficult, 2 questions are easy and 2 questions are moderate	
Algebra - Matrix, trigonometry, sets, complex number, quadratic, inverse trigonometry	1 easy question from trigonometry, 1 tough question from complex number, 1 moderate question from Quadratic & complex number, 2 moderate question from Matrix & determinant, 2 question from Permutation & combination,	1 Easy Question from inverse trigonometry	
Coordinate geometry	1 moderate question from Ellipse, 1 tough question from Straight line, 1 easy question from circles	2 moderate questions from conic sections	

Calculus	1 easy question from limits & differential equation	6 questions from Calculus out of which 1 moderate question is from Application of integrals,1 moderate and 1 tough question from limits, 3 moderate questions from definite integration	
Probability & statistics	2 questions from Probability & statistics out of which one is moderate and the other is tough	1 moderate question from probability	
Vectors, 3D geometry	1 easy question from vectors, 2 questions from 3 D out of which one is easy and the other question is tough	1 moderate question from 3D, 1 moderate question from vectors	

Overall Analysis

Paper Difficulty Compared to Last Year
- Paper was overall easy to moderate - Compared to last year, level of paper was on easier side - Calculation was bit lengthy - In paper 1, 5 questions were tough, 6 questions were easy, 6 questions were moderate. - In paper 2, 2 questions were tough, 5 questions were easy, and rest were moderate - Questions were asked from almost every topic. No questions from binomial theorem - Topic - Functions & relations holds the maximum weightage followed by calculus - Weightage of 12th was slightly more compared to 11th

JEE ADVANCED-5 Year (2023-19) Paper Analysis

Note: Due to unavailability of 2024 paper data we are unable to incorporate the 2024 analysis. As we are coming up with this book before the 2024 result.

Explanation of ★Unique★ Feature

C – Correct, W – Wrong, UA – Unattempted, PC – Partial Correct

C-6.97 W-42.98 UA-33.15 PC-16.9 represents the % of distribution of correct, wrong, unattempted and partial correct responses by students at any specific question in real time.(Data is taken from JEE Advanced website: https://jeeadv.ac.in/reports.html)

Classification helps students understand the varying **levels of difficulty**.

Easy	Medium	Hard
C > 20%	10% < C ≤ 20%	C ≤ 10%

For Example

64. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in R$. Suppose $Q = [q_{ij}]$ is a matrix such that $PQ = kI$, where $k \in R$, $k \neq 0$ and I is the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det(Q) = \frac{k^2}{2}$, then **C-1.96 W-24.58 UA-66.2 PC-7.26 [JEE Adv. 2016]**

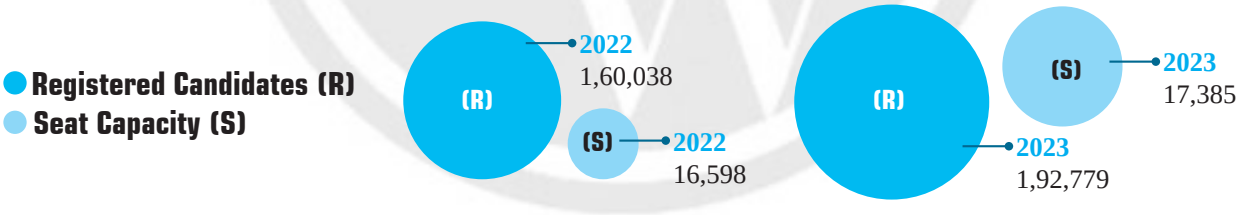
(*a*) $\alpha = 0, k = 8$

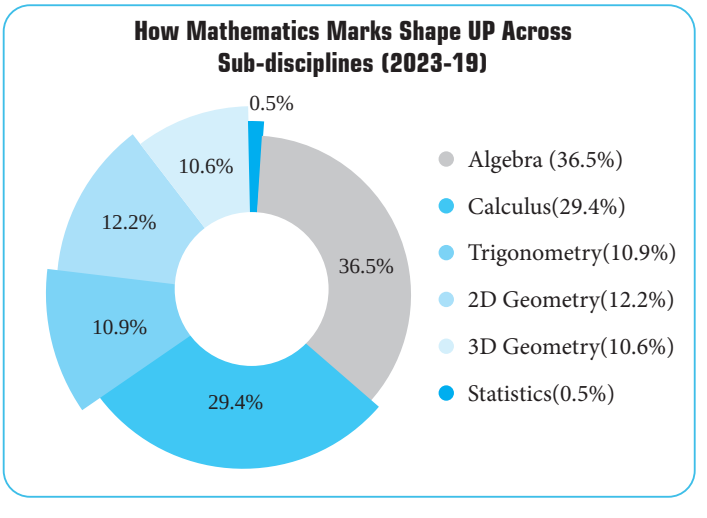
(*b*) $4\alpha - k + 8 = 0$

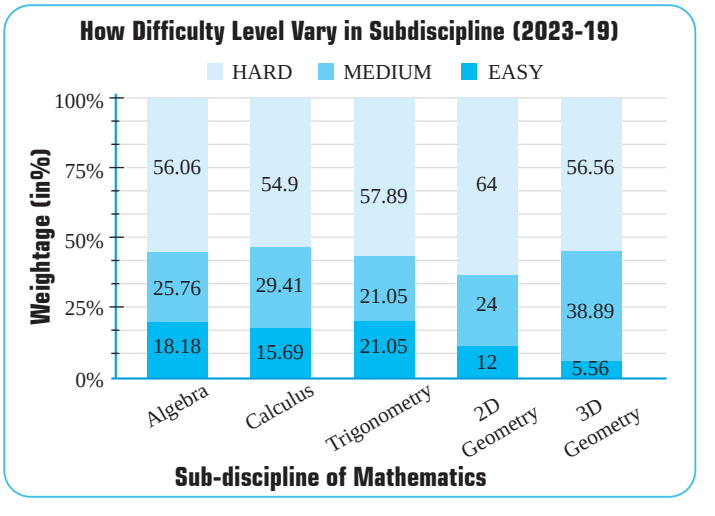
(*c*) $\det(P \operatorname{adj}(Q)) = 2^9$

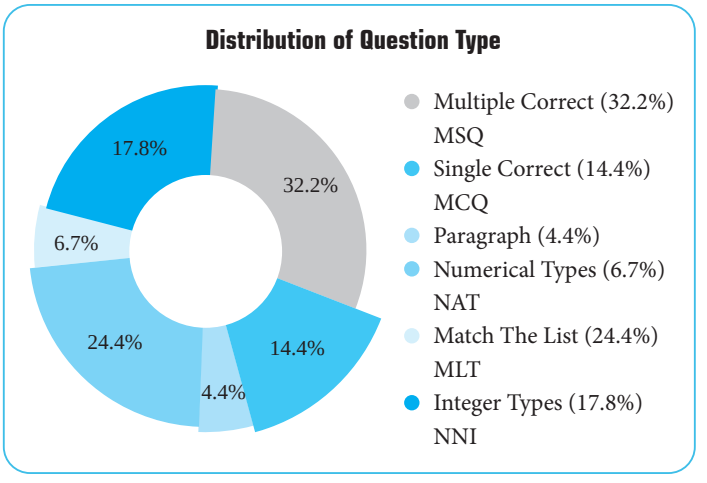
(*d*) $\det(Q \operatorname{adj}(P)) = 2^{13}$

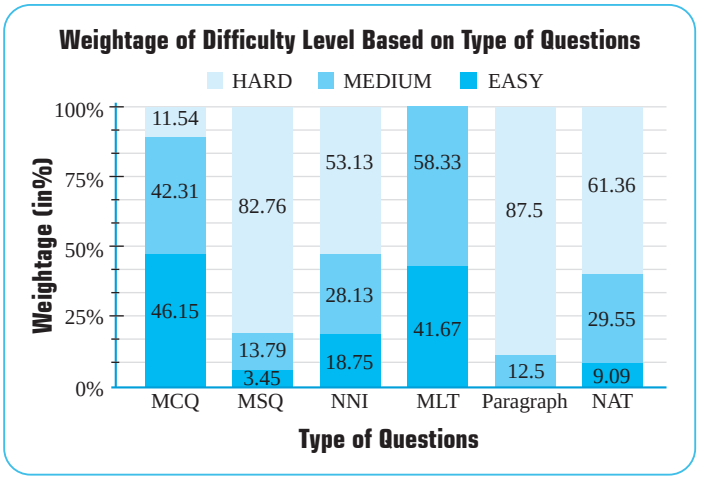
Question is considered HARD as Correct response recorded was less than **10%**

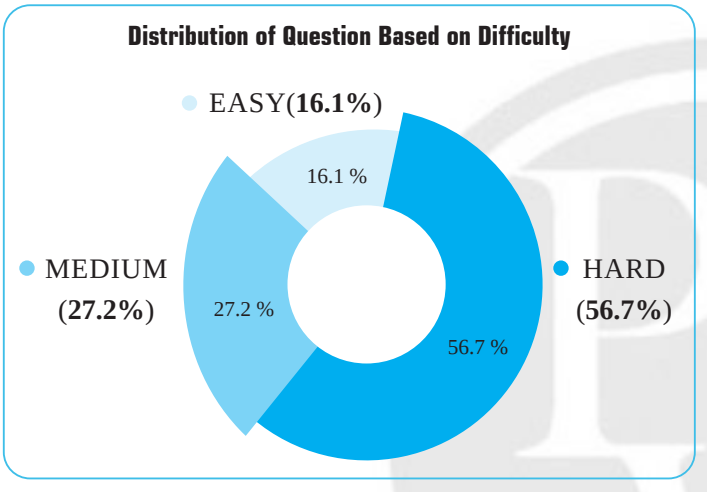


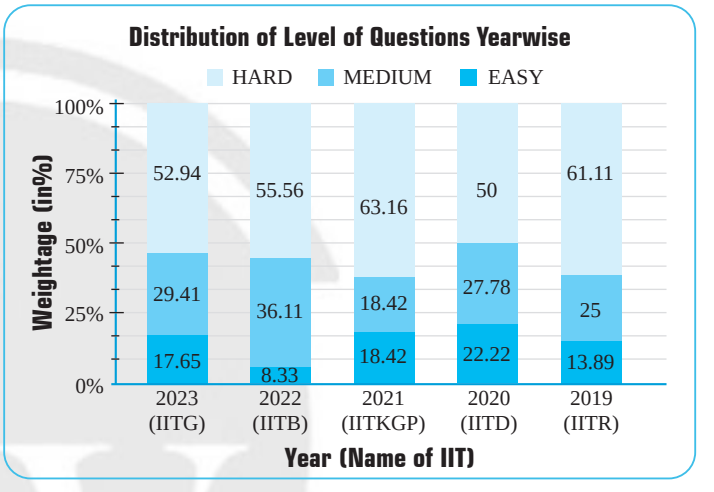












Chapter wise Weightage and count of Difficulty level of JEE Advanced Questions

Chapters Name	EASY	MEDIUM	HARD	Chapterwise Weightage(in %)
Probability	3	4	13	**11.11**
Matrices And Determinants	4	1	11	**8.89**
Application Of Derivatives	2	3	11	**8.89**
Circles	1	3	8	**6.67**
Complex Numbers	0	7	5	**6.67**
Definite Integration	1	4	5	**5.56**
Three Dimensional Geometry	0	4	5	**5**
Vectors	1	3	5	**5**
Solutions Of Triangles	1	0	7	4.44
Limits Of Functions	1	2	4	3.89
Application of Integrals	2	1	3	3.33
Inverse Trigonometric Functions	0	3	3	3.33
Sequence And Series	0	4	2	3.33
Permutations And Combinations	0	1	5	3.33
Ellipse	1	2	2	2.78

Differential Equation	1	2	2	2.78
Trigonometric Equations	3	1	0	2.22
Differentiability And Method Of Differentiation	0	2	2	2.22
Parabola	1	0	2	1.67
Striaght Lines	0	1	2	1.67
Binomial Theorem	1	0	1	1.11
Basic Mathematics	2	0	0	1.11
Hyperbola	0	0	2	1.11
Quadratic Equation	2	0	0	1.11
Functions	1	1	0	1.11
Statistics	1	0	0	0.56
Continuity	0	0	1	0.56
Trigonometric Ratios and Identities	0	0	1	0.56

IIPE
IISC
RGIPT
IISERs
IIST
Non IIT Institutes using JEE Advanced Ranks
APPLY

1 Rank

Candidate Marks (Category-wise)

	Physics 2023		Physics 2022
	110	CRL	96
	109	GEN-EWS	98
	111	OBC-NCL	103
	110	SC	77
	109	ST	71

	Chemistry 2023		Chemistry 2022
	117	CRL	109
	108	GEN-EWS	103
	106	OBC-NCL	110
	97	SC	88
	87	ST	67

	Mathematics 2023		Mathematics 2022
	114	CRL	109
	106	GEN-EWS	91
	103	OBC-NCL	86
	103	SC	78
	69	ST	46

***CRL - Common Rank List**

2024 JEE ADVANCED SOLVED PAPER

Mathematics Paper-1

SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (*a*), (*b*), (*c*) and (*d*). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases.

1. Let $f(x)$ be a continuously differentiable function on the interval $(0, \infty)$ such that $f(1) = 2$ and $\lim_{t\to x}\frac{t^{10}f(x)-x^{10}f(t)}{t^9-x^9}=1$ for each $x > 0$. Then, for all $x > 0$, $f(x)$ is equal to

(*a*) $\frac{31}{11x}-\frac{9}{11}x^{10}$ (*b*) $\frac{9}{11x}+\frac{13}{11}x^{10}$

(*c*) $\frac{-9}{11x}+\frac{31}{11}x^{10}$ (*d*) $\frac{13}{11x}+\frac{9}{11}x^{10}$

2. A student appears for a quiz consisting of only true-false type questions and answers all the questions. The student knows the answers of some questions and guesses the answers for the remaining questions. Whenever the student knows the answer of a question, he gives the correct answer. Assume that the probability of the student giving the correct answer for a question, given that he has guessed it, is $\frac{1}{2}$. Also assume that the probability of the answer for a question being guessed, given that the student's answer is correct, is $\frac{1}{6}$. Then the probability that the student knows the answer of a randomly chosen question is

(*a*) $\frac{1}{12}$ (*b*) $\frac{1}{7}$ (*c*) $\frac{5}{7}$ (*d*) $\frac{5}{12}$

3. Let $\frac{\pi}{2}<x<\pi$ be such that $\cot x=\frac{-5}{\sqrt{11}}$. Then

$\left(\sin\frac{11x}{2}\right)(\sin 6x-\cos 6x)+\left(\cos\frac{11x}{2}\right)(\sin 6x+\cos 6x)$ is equal to

(*a*) $\frac{\sqrt{11}-1}{2\sqrt{3}}$ (*b*) $\frac{\sqrt{11}+1}{2\sqrt{3}}$ (*c*) $\frac{\sqrt{11}+1}{3\sqrt{2}}$ (*d*) $\frac{\sqrt{11}-1}{3\sqrt{2}}$

4. Consider the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$. Let $S(p, q)$ be a point in the first quadrant such that $\frac{p^2}{9}+\frac{q^2}{4}>1$. Two tangents are drawn from S to the ellipse, of which one meets the ellipse at one end point of the minor axis and the other meets the ellipse at a point T in the fourth quadrant. Let R be the vertex of the ellipse with positive x-coordinate and O be the center of the ellipse. If the area of the triangle ΔORT is $\frac{3}{2}$, then which of the following options is correct?

(*a*) $q = 2, p = 3\sqrt{3}$ (*b*) $q = 2, p = 4\sqrt{3}$

(*c*) $q = 1, p = 5\sqrt{3}$ (*d*) $q = 1, p = 6\sqrt{3}$

SECTION 2 (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (*a*), (*b*), (*c*) and (*d*). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –2 In all other cases.

- For example, in a question, if (*a*), (*b*) and (*d*) are the **ONLY** three options corresponding to correct answers, then
choosing ONLY (*a*), (*b*) and (*d*) will get +4 marks;
choosing ONLY (*a*) and (*b*) will get +2 marks;
choosing ONLY (*a*) and (*d*) will get +2 marks;
choosing ONLY (*b*) and (*d*) will get +2 marks;
choosing ONLY (*a*) will get +1 mark;
choosing ONLY (*b*) will get +1 mark;
choosing ONLY (*d*) will get +1 mark;
choosing no option (i.e. the question is unanswered) will get 0 marks; and
choosing any other combination of options will get –2 marks.

5. Let $S=\left\{a+b\sqrt{2}:a,b\in\mathbb{Z}\right\}, T_1=\left\{(-1+\sqrt{2})^n:n\in\mathbb{N}\right\}$, and $T_2=\left\{(1+\sqrt{2})^n:n\in\mathbb{N}\right\}$.

Then which of the following statements is (are) TRUE?

(a) $\mathbb{Z}\cup T_1\cup T_2\subset S$

(b) $T_1\cap\left(0,\frac{1}{2024}\right)=\phi$, where ϕ denotes the empty set.

(c) $T_2\cap(2024,\infty)\neq\phi$

(d) For any given

$a,b\in\mathbb{Z}, \cos\left(\pi\left(a+b\sqrt{2}\right)\right)+i\sin\left(\pi\left(a+b\sqrt{2}\right)\right)\in\mathbb{Z}$ if and only if $b=0$, where $i=\sqrt{-1}$.

6. Let $\mathbb{R}^2$ denote $\mathbb{R}\times\mathbb{R}$. Let

$$S=\left\{\begin{array}{l}(a,b,c):a,b,c\in\mathbb{R} \text{ and } ax^2+2bxy+cy^2>0\\ \text{for all } (x,y)\in\mathbb{R}^2-\{(0,0)\}\end{array}\right\}$$

Then which of the following statements is (are) TRUE?

(a) $\left(2,\frac{7}{2},6\right)\in S$

(b) If $\left(3,b,\frac{1}{12}\right)\in S$, then $|2b|<1$.

(c) For any given $(a, b, c)\in S$, the system of linear equations

$ax+by=1$

$bx+cy=-1$

has a unique solution.

(d) For any given $(a, b, c)\in S$, the system of linear equations

$(a+1)x+by=0$

$bx+(c+1)y=0$

has a unique solution.

7. Let $\mathbb{R}^3$ denote the three-dimensional space. Take two points $P=(1, 2, 3)$ and $Q=(4, 2, 7)$. Let $dist(X, Y)$ denote the distance between two points X and Y in $\mathbb{R}^3$. Let

$$S=\left\{X\in\mathbb{R}^3:(dist(X,P))^2-(dist(X,Q)^2)=50\right\} \text{ and}$$

$$T=\left\{Y\in\mathbb{R}^3:(dist(Y,Q))^2-(dist(Y,P))^2=50\right\}$$

Then which of the following statements is (are) TRUE?

(a) There is a triangle whose area is 1 and all of whose vertices are from S.

(b) There are two distinct points L and M in T such that each point on the line segment LM is also in T.

(c) There are infinitely many rectangles of perimeter 48, two of whose vertices are from S and the other two vertices are from T.

(d) There is a square of perimeter 48, two of whose vertices are from S and the other two vertices are from T.

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If **ONLY** the correct integer is entered;

Zero Marks : 0 In all other cases.

8. Let $a=3\sqrt{2}$ and $b=\frac{1}{5^{1/6}\sqrt{6}}$. If $x,y\in\mathbb{R}$ are such that

$$3x+2y=\log_a(18)^{\frac{5}{4}} \text{ and } 2x-y=\log_b\left(\sqrt{1080}\right)$$

then $4x+5y$ is equal to _____.

9. Let $f(x)=x^4+ax^3+bx^2+c$ be a polynomial with real coefficients such that $f(1)=-9$. Suppose that $i\sqrt{3}$ is a root of the equation $4x^3+3ax^2+2bx=0$, where $i=\sqrt{-1}$. If $\alpha_1, \alpha_2, \alpha_3$, and α_4 are all the roots of the equation $f(x)=0$, then $|\alpha_1|^2+|\alpha_2|^2+|\alpha_3|^2+|\alpha_4|^2$ is equal _____.

10. Let $S=\left\{A=\begin{pmatrix}0&1&c\\1&a&d\\1&b&e\end{pmatrix}:a,b,c,d,e\in\{0,1\} \text{ and } |A|\in\{-1,1\}\right\}$,

where $|A|$ denotes the determinant of A. Then the number of elements in S is_____.

11. A group of 9 students, $S_1, S_2 ..., S_9$, is to be divided to form three teams X, Y and Z of sizes 2, 3, and 4, respectively. Suppose that S_1 cannot be selected for the team X, and S_2 cannot be selected for the team Y. Then the number of ways to form such teams, is _____.

12. Let $\overrightarrow{OP}=\frac{\alpha-1}{\alpha}\hat{i}+\hat{j}+\hat{k}$, $\overrightarrow{OQ}=\hat{i}+\frac{\beta-1}{\beta}\hat{j}+\hat{k}$ and $\overrightarrow{OR}=\hat{i}+\hat{j}+\frac{1}{2}\hat{k}$

be three vectors, where $\alpha,\beta\in\mathbb{R}-\{0\}$. and O denotes the origin.

If $(\overrightarrow{OP}\times\overrightarrow{OQ})\cdot\overrightarrow{OR}=0$ and the point $(\alpha, \beta, 2)$ lies on the plane $3x+3y-z+l=0$, then the value of l is _____.

13. Let X be a random variable, and let $P(X=x)$ denote the probability that X takes the value x. Suppose that the points $(x, P(X=x))$, $x=0, 1, 2, 3, 4$ lie on a fixed straight line in the xy-plane, and $P(X=x)=0$ for all $x\in\mathbb{R}-\{0,1,2,3,4\}$. If the mean of X is $\frac{5}{2}$, and the variance of X is α, then the value of 24α is ____.

SECTION 4 (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has Five entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : –1 In all other cases.

14. Let α and b be the distinct roots of the equation $x^2 + x - 1 = 0$. Consider the set $T = \{1, \alpha, \beta\}$. For a 3×3 matrix $M = (a_{ij})_{3\times3}$, define $R_i = a_{i1} + a_{i2} + a_{i3}$ and $C_j = a_{1j} + a_{2j} + a_{3j}$ for $i = 1, 2, 3$ and $j = 1, 2, 3$

Match each entry in **List-I** to the correct entry in **List-II**.

	List-I		List-II
(P)	The number of matrices $M = (a_{ij})_{3\times3}$ with all entries in T such that $R_i = C_j = 0$ for all i, j, is	(1)	1
(Q)	The number of symmetric matrices $M = (a_{ij})_{3\times3}$ with all entries in T such that $C_j = 0$ for all j, is	(2)	12
(R)	Let $M = (a_{ij})_{3\times3}$ be a skew symmetric matrix such that $a_{ij} \in T$ for $i > j$. Then the number of elements in the set is $\left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix} : x, y, z \in \mathbb{R}, M\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_{12} \\ 0 \\ -a_{23} \end{pmatrix}\right\}$	(3)	infinite
(S)	Let $M = (a_{ij})_{3\times3}$ be a matrix with all entries in T such that $R_i = 0$ for all i. Then the absolute value of the determinant of M is	(4)	6
		(5)	0

The correct option is

(*a*) (P) → (4) (Q) → (2) (R) → (5) (S) → (1)
(*b*) (P) → (2) (Q) → (4) (R) → (1) (S) → (5)
(*c*) (P) → (2) (Q) → (4) (R) → (3) (S) → (5)
(*d*) (P) → (1) (Q) → (5) (R) → (3) (S) → (4)

15. Let the straight line $y = 2x$ touch a circle with center $(0, \alpha)$, $\alpha > 0$, and radius r at a point A_1. Let B_1 be the point on the circle such that the line segment A_1B_1 is a diameter of the circle. Let $\alpha + r = 5 + \sqrt{5}$.

Match each entry in **List-I** to the correct entry in **List-II**.

	List-I		List-II
(P)	α equals	(1)	(–2, 4)
(Q)	r equals	(2)	$\sqrt{5}$
(R)	A_1 equals	(3)	(–2, 6)
(S)	B_1 equals	(4)	5
		(5)	(2, 4)

The correct option is

(*a*) (P) → (4) (Q) → (2) (R) → (1) (S) → (3)
(*b*) (P) → (2) (Q) → (4) (R) → (1) (S) → (3)
(*c*) (P) → (4) (Q) → (2) (R) → (5) (S) → (3)
(*d*) (P) → (2) (Q) → (4) (R) → (3) (S) → (5)

16. Let $\gamma \in \mathbb{R}$ be such that the lines $L_1 : \frac{x+11}{1} = \frac{y+21}{2} = \frac{z+29}{3}$ and $L_2 : \frac{x+16}{3} = \frac{y+11}{2} = \frac{z+4}{\gamma}$ intersect. Let R_1 be the point of intersection of L_1 and L_2. Let $O = (0, 0, 0)$, and $\hat{n}$ denote a unit normal vector to the plane containing both the lines L_1 and L_2.

Match each entry in **List-I** to the correct entry in **List-II**.

	List-I		List-II
(P)	γ equals	(1)	$-\hat{i} - \hat{j} + \hat{k}$
(Q)	A possible choice for $\hat{n}$ is	(2)	$\sqrt{\frac{3}{2}}$
(R)	$\overrightarrow{OR_1}$ equals	(3)	1
(S)	A possible value of $\overrightarrow{OR_1} \cdot \hat{n}$ is	(4)	$\frac{1}{\sqrt{6}}\hat{i} - \frac{2}{\sqrt{6}}\hat{j} + \frac{1}{\sqrt{6}}\hat{k}$
		(5)	$\sqrt{\frac{2}{3}}$

The correct option is

(*a*) (P) → (3) (Q) → (4) (R) → (1) (S) → (2)
(*b*) (P) → (5) (Q) → (4) (R) → (1) (S) → (2)
(*c*) (P) → (3) (Q) → (4) (R) → (1) (S) → (5)
(*d*) (P) → (3) (Q) → (1) (R) → (4) (S) → (5)

17. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be functions defined by

$$f(x) = \begin{cases} x|x|\sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0, \end{cases} \text{ and } g(x) = \begin{cases} 1 - 2x, & 0 \le x \le \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$$

Let $a, b, c, d \in \mathbb{R}$. Define the function $h : \mathbb{R} \to \mathbb{R}$ by

$$h(x) = af(x) + b\left(g(x) + g\left(\frac{1}{2} - x\right)\right) + c(x - g(x)) + dg(x), x \in \mathbb{R}$$

Match each entry in **List-I** to the correct entry in **List-II**.

	List-I		List-II
(P)	If $a = 0$, $b = 1$, $c = 0$, and $d = 0$, then	(1)	h is one-one.
(Q)	If $a = 1$, $b = 0$, $c = 0$, and $d = 0$, then	(2)	h is onto.
(R)	If $a = 0$, $b = 0$, $c = 1$, and $d = 0$, then	(3)	h is differentiable on $\mathbb{R}$.
(S)	If $a = 0$, $b = 0$, $c = 0$, and $d = 1$, then	(4)	the range of h is [0, 1].
		(5)	the range of h is {0, 1}.

The correct option is

(*a*) (P) → (4) (Q) → (3) (R) → (1) (S) → (2)
(*b*) (P) → (5) (Q) → (2) (R) → (4) (S) → (3)
(*c*) (P) → (5) (Q) → (3) (R) → (2) (S) → (4)
(*d*) (P) → (4) (Q) → (2) (R) → (1) (S) → (3)

Solutions Paper-1

1. (*b*) $\lim_{t\to x}\dfrac{t^{10}f(x)-x^{10}f(t)}{t^9-x^9}=1$

$\dfrac{0}{0}$ form, use L' Hospital

$\lim_{t\to x}\dfrac{10t^9f(x)-f'(t)x^{10}}{9t^8}=1$

$\Rightarrow 10x^9f(x)-f'(x)x^{10}=9x^8$

$\Rightarrow f'(x)-\dfrac{10}{x}f(x)=-\dfrac{9}{x^2}$

It is of form $\dfrac{dx}{dy}+Rx=S$

$\text{I.F}=e^{-\int\frac{10}{x}dx}=\dfrac{1}{x^{10}}$

Let $f(x)=y$

$\dfrac{y}{x^{10}}=\int-\dfrac{9}{x^{10}}\times\dfrac{1}{x^2}dx$

$\Rightarrow\dfrac{y}{x^{10}}=\dfrac{9}{11}x^{-11}+C$

$\because y(1)=2\Rightarrow C=\dfrac{13}{11}$

$\Rightarrow y=\dfrac{9}{11x}+\dfrac{13}{11}x^{10}$

2. (*c*) Let P(knows answer) = p

P (guesses) = $1-p$

$P\left(\dfrac{\text{correct ans}}{\text{guessed}}\right)=\dfrac{1}{2}$

$P\left(\dfrac{\text{guessed}}{\text{correct answer}}\right)=$

$$\frac{P(\text{ guessed})P\left(\dfrac{\text{correct ans}}{\text{guessed}}\right)}{P(\text{ guessed})P\left(\dfrac{\text{correct ans}}{\text{guessed}}\right)+P(\text{ knows})P\left(\dfrac{\text{correct ans}}{\text{knows}}\right)}$$

$\Rightarrow\dfrac{1}{6}=\dfrac{(1-p)\left(\frac{1}{2}\right)}{(1-p)\left(\frac{1}{2}\right)+p(1)}\Rightarrow k=\dfrac{5}{7}$

3. (*b*) Let

$F=\sin 6x\cos\dfrac{11x}{2}-\cos 6x\sin\dfrac{11x}{2}$

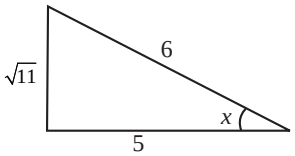
$+\cos 6x\cos\dfrac{11x}{2}+\sin 6x\sin\dfrac{11x}{2}$

Using, $\sin(A-B)=\sin A\cos B-\cos A\sin B$ and $\cos(A-B)=\cos A\cos B+\sin A\sin B$

$F=\sin\dfrac{x}{2}+\cos\dfrac{x}{2}$

Now, $F^2=1+\sin x$ $\left[\because \cot x=\dfrac{-5}{\sqrt{11}}\right]$

$=1+\dfrac{\sqrt{11}}{6}$

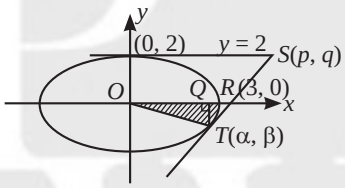


$F^2=\dfrac{12+2\sqrt{11}}{12}=\dfrac{11+1+2\sqrt{11}}{12}$

$=\dfrac{\left(\sqrt{11}+1\right)^2}{12}$

$\therefore F=\dfrac{\sqrt{11}+1}{2\sqrt{3}}$

4. (*a*)



Given, $\dfrac{p^2}{q}+\dfrac{q^2}{4}>1$

$\Rightarrow (p, q)$ outside the ellipse

Area $(ORT)=\dfrac{3}{2}$

$\Rightarrow\left|\dfrac{1}{2}\times OR\times QT\right|=\dfrac{3}{2}$

$\Rightarrow\left|\dfrac{1}{2}\times 3\times\beta\right|=\dfrac{3}{2}$

$\Rightarrow\beta=\pm 1\Rightarrow\beta=-1$

[$\because$ in 4th quadrant]

$\therefore\dfrac{\alpha^2}{9}+\dfrac{\beta^2}{4}=1$

$\Rightarrow\dfrac{\alpha^2}{9}=1-\dfrac{1}{4}=\dfrac{3}{4}$

$\Rightarrow\alpha^2=\dfrac{27}{4}\Rightarrow\alpha=\dfrac{3\sqrt{3}}{2}$

Since, one of the tangent touches minor axis at one of its end i.e., tangent is parallel to x axis.

$\therefore q=2$

Tangent at T, $T=0$

$\dfrac{x\cdot\frac{3\sqrt{3}}{2}}{9}+\dfrac{y(-1)}{4}=1\Big|_{(p,2)}$

$\Rightarrow\dfrac{p\sqrt{3}}{6}-\dfrac{1}{2}=1\Rightarrow\dfrac{p\sqrt{3}}{6}=\dfrac{3}{2}$

$\Rightarrow p=3\sqrt{3}$

$\therefore p=3\sqrt{3}, q=2$

5. (*a, c, d*) (*a*) $S=\left\{a+b\sqrt{2}:a,b\in Z\right\}$, If $b=0$, then $Z\subset S$

$T_1=\left\{(-1+\sqrt{2})^n:n\in N\right\}$

Hence, T_1 is of form $(-1+\sqrt{2})$, $\left(3-2\sqrt{2}\right)$ ..... So on.

and $T_2=\left\{(1+\sqrt{2})^n:n\in N\right\}$

Hence, T_2 is of form $\left(1+\sqrt{2}\right),\left(3+2\sqrt{2}\right)$ .... So on.

$\therefore n\in N$ elements of T_1 and T_2 are of the form $a+b\sqrt{2}$

Hence $Z\cup T_1\cup T_2\subset S$

(*b*) Now, $-1+\sqrt{2}<1$ and its higher powers decreases

$\Rightarrow \text{In}\,(-1+\sqrt{2})^n<1$ some values in between $\left(0,\dfrac{1}{2024}\right)$ can lie for some higher n.

$\therefore T_1\cap\left(0,\dfrac{1}{2024}\right)\neq\phi$

(*c*) $1+\sqrt{2}>1$ and its higher power increases

$\Rightarrow \text{In}\,(1+\sqrt{2})^n$ some values in between $(2024,\infty)$ can lie for some higher n.

$\therefore T_2\cap(2024,\infty)\neq\phi$

(*d*)

$\cos\pi\left(a+b\sqrt{2}\right)+i\sin\pi\left(a+b\sqrt{2}\right)\in Z$

if $a+b\sqrt{2}$ is an integer $\Rightarrow b=0$

6. (*b, c, d*) $ax^2+2bxy+cy^2>0\ \forall$ $y,x\in\mathbb{R}^2-\{(0,0)\}$

Divide by x^2

$\Rightarrow c\left(\dfrac{y}{x}\right)^2+2b\left(\dfrac{y}{x}\right)+a>0$

Let, $\dfrac{y}{x}=m$

$\Rightarrow cm^2+2bm+a>0\ \forall\ m\in R$

$\Rightarrow c>0, D<0$

$\therefore 4b^2-4ac<0$

$\Rightarrow b^2<ac$

(*a*) If $\left(2,\frac{7}{2},6\right)\in S$

$\Rightarrow\left(\frac{7}{2}\right)^2>2\times6$

$\therefore$ option (*a*) is incorrect

(*b*) If $\left(3,b,\frac{1}{12}\right)\in S$

$\Rightarrow b^2<3\cdot\frac{1}{12}\Rightarrow 4b^2<1$

$\Rightarrow |2b|<1$

$\therefore$ option (*b*) is correct

(*c*) $ax+by=1$

$bx+cy=-1$

$D=\begin{vmatrix}a & b\\ b & c\end{vmatrix}=ac-b^2\neq0$

$\therefore$ unique solution

$\Rightarrow$ Option (*c*) is correct.

(*d*) $(a+1)x+by=0$

$bx+(c+1)y=0$

$D=\begin{vmatrix}(a+1) & b\\ b & (c+1)\end{vmatrix}$

$=(a+1)(c+1)-b^2$

$\Rightarrow ac-b^2+a+c+1$

$b^2<ac\Rightarrow ac$ is +ve

$\Rightarrow a$ and c both are positive then $(ac-b^2)+a+c+1>0$

$\therefore$ unique solution $\Rightarrow$ option (*d*) is correct

7. (*a*, *b*, *c*, *d*) *S*: $\{((x-1)^2+(y-2)^2(z-3)^2)-((x-4)^2+(y-2)^2+(z-7)^2)=50\}$

$\Rightarrow S$: $\{6x+8z-105=0\}$

Now, T: $\{((x-4)^2+(y-2)^2(z-7)^2)-((x-1)^2+(y-2)^2+(z-3)^2)=50\}$

S: $\{(x^2+1-2x+y^2+4-4y+z^2+9-6z-x^2-16+8x-y^2-4+4y-z^2-49+14z=50\}$

$\Rightarrow$ T = $\{6x+8z-5=0\}$

(*a*) S represents a plane. So it will contain a triangle of area 1. So (*a*) is correct.

(*b*) *T* represents a plane. So (*b*) is also correct.

(*c*) S and T are two parallel planes at a distance of 10 units from each other.

$\therefore$ (*c*) is also correct

(*d*) Perimeter of square = 48 $\Rightarrow$ side = 12, possible if square is slant

8. [8] $a=3\sqrt{2}\Rightarrow a^2=18$

$\therefore \log_{\sqrt{18}}18^{\frac{5}{4}}=\frac{\frac{5}{4}}{\frac{1}{2}}=\frac{5}{2}\Rightarrow 3x+2y=\frac{5}{2}$ … (*i*)

$b=5^{-\frac{1}{6}\cdot6^{-\frac{1}{2}}}=5^{-\frac{1}{6}\cdot6^{-\frac{3}{6}}}$

$=\left(5\times6^3\right)^{-\frac{1}{6}}=1080^{-\frac{1}{6}}$

$\therefore \log_{(1080)^{-\frac{1}{6}}}\sqrt{1080}=\frac{\frac{1}{2}}{-\frac{1}{6}}=-3$

$2x-y=-3$ … (*ii*)

$\Rightarrow$ Solving (*i*) & (*ii*)

$\Rightarrow x=\frac{-1}{2},y=2\Rightarrow 4x+5y=8$

9. [20]

$\because f(1)=-9\Rightarrow 1+a+b+c=-9$

$\Rightarrow a+b+c=-10$ … (*i*)

$4x^3+3ax^2+2bx=0$

$x(4x^2+3ax+26)=0$

$\Rightarrow x=0$, or $4x^2+3ax+2b=0$ … (*ii*)

Given, $\sqrt{3}i$ and $-\sqrt{3}i$ are roots of eqn (*ii*)

$\therefore$ Sum of roots = $\sqrt{3}i-\sqrt{3}i=\frac{-3a}{4}$

$\Rightarrow a=0$

Also, product of roots

$=\sqrt{3}i\left(-\sqrt{3}i\right)=\frac{26}{4}\Rightarrow b=6$

$\Rightarrow a=0,b=6,c=-16$ [using (*ii*)]

Now, $f(x)=0\Rightarrow x^4+6x^2-16=0$

$\Rightarrow x^2=\frac{-6\pm\sqrt{36+64}}{2}=-3\pm5=2,-8$

$x=-\sqrt{2},+\sqrt{2},-2\sqrt{2}i,2\sqrt{2}i$

$\Rightarrow|\alpha_1|^2+|\alpha_2|^2+|\alpha_3|^2+|\alpha_4|^2=20$

10. [16] $|A|=-(e-d)+c(b-a)=\pm1$

Case (i):

$c=0\Rightarrow(e,d)=(1,0),(0,1)\to2$ ways

b and *a* can be each 2 ways

$\Rightarrow$ Total = 2 × 2 × 2 = 8 ways

Case (ii): $c\neq0\Rightarrow c=1$

$\Rightarrow d-e+b-a=\pm1$

$\left.\begin{matrix}1&0&0&0\\1&1&1&0\\0&0&1&0\\1&0&1&1\end{matrix}\right\}\to4\times2=8$ ways

$\therefore$ Total ways = 16

11. [665] The number of ways

$\frac{9!}{2!3!4!}-(n(s_1\in X)+n(s_2\in Y)-n(s_1\in X$ and $s_2\in Y))$

$=\frac{9!}{2!3!4!}-\left(\frac{8!}{1!3!4!}+\frac{8!}{2!2!4!}-\frac{7!}{1!2!4!}\right)=665$

12. [5] Given, $(\overrightarrow{OP}\times\overrightarrow{OQ})\cdot\overrightarrow{OR}=0$

$\therefore\begin{vmatrix}\frac{\alpha-1}{\alpha} & 1 & 1\\ 1 & \frac{\beta-1}{\beta} & 1\\ 1 & 1 & \frac{1}{2}\end{vmatrix}=0$

$\Rightarrow\frac{\alpha-1}{\alpha}\left(\frac{\beta-1}{2\beta}-1\right)-1\left(\frac{1}{2}-1\right)+1\left(1-\frac{\beta-1}{\beta}\right)=0$

$\Rightarrow\frac{\alpha-1}{\alpha}\left(\frac{-\beta-1}{2\beta}\right)+\frac{1}{2}+\frac{1}{\beta}=0$

$\Rightarrow\frac{1}{2}+\frac{1}{\beta}=\frac{(\alpha-1)}{\alpha}\left(\frac{\beta+1}{2\beta}\right)$

$\Rightarrow\frac{\beta+2}{2\beta}=\frac{\alpha\beta+\alpha-\beta-1}{2\alpha\beta}$

$\Rightarrow\alpha\beta+2\alpha=\alpha\beta+\alpha-\beta-1$

$\Rightarrow\alpha+\beta+1=0$ … (*i*)

Since (α, β, 2) lies on $3x+3y-z+l=0$

$\therefore 3(\alpha+\beta)-2+l=0$

From eq. (*i*)

$\Rightarrow -3-2+l=0\Rightarrow l=5$

13. [42] Given, $\sum_{x=0}^{4}xP(x)=\frac{5}{2}$

$(0,P(0)),(1,P(1)),(2,P(2)),(3,P(3)),(4,P(4))$

$K=P(1)-P(0)=P(2)-P(1)=P(3)-P(2)$

$=P(4)-P(3)$

$P(1)=K+P(0)$, $P(2)=2K+P(0)$,

$P(3)=3K+P(0)$ and $P(4)=4K+P(0)$

$\therefore P(0)+P(1)+P(2)+P(3)+P(4)=1$

$\Rightarrow 5P(0)+10K=1$

$\sum_{x=0}^{4}xP(x)=K+P(0)+4K+2P(0)+9K+3P(0)+16K+4P(0)=\frac{5}{2}$

$\Rightarrow 30K+10P(0)=\frac{5}{2}\Rightarrow 10K=\frac{1}{2}\Rightarrow K=\frac{1}{20}$

$P(0)=\frac{1}{10}$

$P(1)=\frac{3}{20},P(2)=\frac{4}{20},P(3)=\frac{5}{20},P(4)=\frac{6}{20}$

$\sum_{x=0}^{4}x^2P(x)=8$

$\therefore$ Variance = $8-\frac{25}{4}=\frac{32-25}{4}=\frac{7}{4}$

$\therefore 24\alpha=\frac{24\times7}{4}=42$

14. (*c*) since of α and β are roots $x^2+x-1=0$

$\therefore\alpha+\beta=-1$ and $\alpha\beta=-1$

Given, $T=\{1,\alpha,\beta\}$

$R_i=a_{i1}+a_{i2}+a_{i3}$ and $C_j=a_{1j}+a_{2j}+a_{3j}$

(*P*) $R_i=C_j=0\ \forall\ i,j$

$\therefore$ The number of matrices

$=\underline{|3}\times2\times1=12$

Number of ways to arrange 1, α, β in R_3

Number of ways to arrange 1, α, β in R_1

Number of ways to arrange 1, α, β in R_2

(*Q*) Given $C_j=0$ for all *j*

Number of symmetric matrices

$=\underline{|3}\times1=6\begin{bmatrix}1&\alpha&\beta\\ \alpha&\beta&1\\ \beta&1&\alpha\end{bmatrix}$

(R) Since M is a skew symmetric of order 3×3

$\therefore |M| = 0, a_{ij} \in T$ for $i > j$

Given, $M\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_{12} \\ 0 \\ -a_{23} \end{pmatrix}$

$$\begin{bmatrix} 0 & -a_{21} & -a_{31} \\ a_{21} & 0 & -a_{32} \\ a_{31} & a_{32} & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a_{12} \\ 0 \\ -a_{23} \end{bmatrix}$$

As $x, y, z \in R$ and a_{12} and $a_{23} \in R$ and $|M| = 0$

$\therefore$ System has infinite solutions

(S) Given, $R_i = 0$ for all i

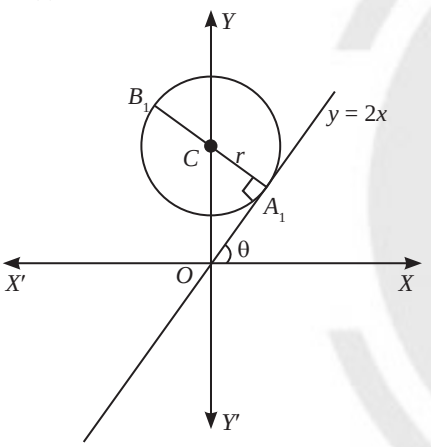
$$\therefore M = \begin{bmatrix} 1 & \alpha & \beta \\ \alpha & \beta & 1 \\ \beta & 1 & \alpha \end{bmatrix}$$

Applying $C_3 \to C_1 + C_2 + C_3$

$$\therefore |M| = \begin{vmatrix} 1 & \alpha & \alpha+\beta+1 \\ \alpha & \beta & \alpha+\beta+1 \\ \gamma & 1 & \alpha+\beta+1 \end{vmatrix} = 0$$

$(P) \to (2), (Q) \to (4), (R) \to (3), (S) \to (5)$

15. (*c*)



Given, $\alpha + r = 5 + \sqrt{5}$ … (*i*)

Since line $y = 2x$ is tangent to the circle

$$\therefore \left|\frac{0-\alpha}{\sqrt{4+1}}\right| = r$$

$\Rightarrow |-\alpha| = r\sqrt{5}$

$\Rightarrow \alpha = r\sqrt{5}$ as $\alpha > 0$

From equation (*i*) $r\sqrt{5} + r = 5 + \sqrt{5}$.

$\Rightarrow r = \sqrt{5}$

And $\alpha = r\sqrt{5} = 5$

Centre $C(0, 5)$

$OC = 5$

$A_1C = \sqrt{5}$

$\therefore OA_1 = \sqrt{25-5} = \sqrt{20} = 2\sqrt{5}$

$\therefore \tan\theta = 2$

Slope of the given line is 2

$$\therefore \cos\theta = \frac{1}{\sqrt{5}} \text{ and } \sin\theta = \frac{2}{\sqrt{5}}$$

Now, $A_1(0 + OA_1 \cos\theta, 0 + OA_1 \sin\theta)$

$$A_1\left(2\sqrt{5}\times\frac{1}{\sqrt{5}}, 2\sqrt{5}\times\frac{2}{\sqrt{5}}\right)$$

$A_1(2, 4)$

Let $B_1(x_1, y_1)$

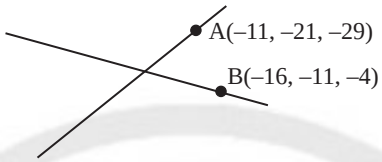
$$\therefore \frac{x_1+2}{2} = 0 \text{ and } \frac{y_1+4}{2} = 5$$

$x_1 = -2$ and $y_1 = 6$

$\therefore B_1(-2, 6)$

16. (*c*) The vector parallel to the line L_1 (say)

$= \hat{i} + 2\hat{j} + 3\hat{k}$



Normal vector of plane $(\vec{n})$ containing L_1 and L_2 will be perpendicular to both $\vec{b_1}$ and $\overrightarrow{AB}$

$\Rightarrow \vec{n} = p(\overrightarrow{AB}\times\vec{n})$

$= (5\hat{i} - 10\hat{j} - 25\hat{k})\times(\hat{i} + 2\hat{j} + 3\hat{k})$

$= (20\hat{i} - 40\hat{j} + 20\hat{k})$

$$\Rightarrow \hat{n} = \frac{1}{\sqrt{6}}\hat{i} - \frac{2}{\sqrt{6}}\hat{j} + \frac{1}{\sqrt{6}}\hat{k}$$

Now, the vector parallel to L_2 (say $\vec{b_2}$) is perpendicular to $\vec{n}$.

$\vec{b_2}\cdot\vec{n} = 0$

$\Rightarrow (3\hat{i} + 2\hat{j} + \gamma\hat{k})\cdot(20\hat{i} - 40\hat{j} + 20\hat{k}) = 0$

$\Rightarrow \gamma = 1$

Now, for point of intersection

$$\frac{x+11}{1} = \frac{y+21}{2} = \frac{z+29}{3} = \lambda \text{ and}$$

$$\frac{x+16}{3} = \frac{y+11}{2} = \frac{z+4}{\gamma} = \mu$$

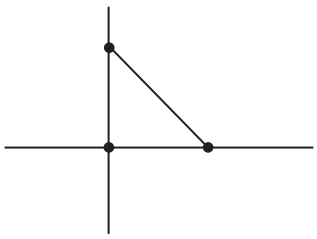
Comparing x and y coordinates,

$-11 + \lambda = -16 + 3\mu$ and $-21 + 2\lambda = -11 + 2\mu$

$\Rightarrow \lambda = 10, \mu = 5$

$$\Rightarrow \overrightarrow{OR_1} : (-\hat{i} - \hat{j} + \hat{k}) \text{ and } \overrightarrow{OR}\cdot\hat{n} = \sqrt{\frac{2}{3}}$$

17. (*c*) Given, $g(x) = \begin{cases} 1-2x, & 0 \le x \le \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$



Now, $g\left(\frac{1}{2} - x\right)$

$$= \begin{cases} 1-2\left(\frac{1}{2}-x\right), & 0 \le \frac{1}{2}-x \le \frac{1}{2} \\ 0, & \text{otherwsie} \end{cases}$$

$$= \begin{cases} 2x, 0 \le x \le \frac{1}{2} \\ 0, \text{ otherwise} \end{cases}$$

$$\therefore g(x) + g\left(\frac{1}{2}-x\right) = \begin{cases} 1-2x+2x, 0 \le x \le \frac{1}{2} \\ 0, \text{ otherwise} \end{cases}$$

$$= \begin{cases} 1, 0 \le x \le \frac{1}{2} \\ 0, \text{ otherwise} \end{cases}$$

Now, option (*P*), at $b = 1$

$h(x) = g(x) + g\left(\frac{1}{2} - x\right)$ has range $\{0, 1\}$

$(P) \to (5)$

Option (*Q*) at $a = 1$, $h(x) = f(x)$

$$f'(0^+) = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h}$$

$$= \lim_{h\to 0}\frac{f(h)-f(0)}{h} = \lim_{h\to 0}\frac{h^2\sin\frac{1}{h}}{h}$$

$$f'(0^-) = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}$$

$$= \lim_{h\to 0}\frac{f(-h)-f(0)}{-h} = \lim_{h\to 0} h\sin h = 0$$

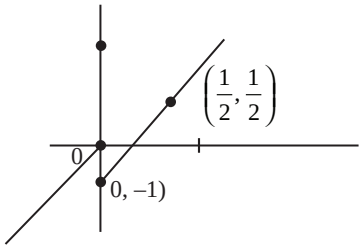
$\Rightarrow f'(0^+) = f'(0^-)$

$\Rightarrow h(x) = f(x)$ is differentiable at $x = 0$ and all other points $f(x) = h(x)$ is differentiable as product of two differentiable functions.

$(Q) \to 3$

Option (*R*)

$$h(x) = x - g(x) = \begin{cases} 3x-1, & 0 \le x \le \frac{1}{2} \\ x, & \text{otherwise} \end{cases}$$



by graph $h(x)$ has range R and onto

$(R) \to (2)$

(*S*) at $d = 1$, $h(x) = g(x)$ has range $[0, 1]$

$(S) \to (4)$

1 QUADRATIC EQUATIONS

JEE-Advanced

Nature of Roots, Relation Between Roots and Coefficients

Single Correct

1. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020 = 0$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020 = 0$. Then the value of $ac(a-c) + ad(a-d) + bc(b-c) + bd(b-d)$ is
C-41.95 W-32.92 UA-25.13 [JEE Adv. 2020]

(*a*) 0 (*b*) 8000
(*c*) 8080 (*d*) 16000

2. Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$, and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals
C-18.36 W-48.75 UA-32.89 [JEE Adv. 2016]

(*a*) $2(\sec\theta - \tan\theta)$ (*b*) $2\sec\theta$
(*c*) $-2\tan\theta$ (*d*) 0

3. In the quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then, the equation $p[p(x)] = 0$ has
C-14.2 W-56.37 UA-29.43 [JEE Adv. 2014]

(*a*) Only purely imaginary roots
(*b*) All real roots
(*c*) Two real and two purely imaginary roots
(*d*) Neither real nor purely imaginary roots

4. Let p and q be real numbers such that $p \neq 0, p^3 \neq q$ and $p^3 \neq -q$. If α and β are non-zero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is **[IIT-JEE 2010]**

(*a*) $(p^3+q)x^2 - (p^3+2q)x + (p^3+q) = 0$
(*b*) $(p^3+q)x^2 - (p^3-2q)x + (p^3+q) = 0$
(*c*) $(p^3-q)x^2 - (5p^3-2q)x + (p^3-q) = 0$
(*d*) $(p^3-q)x^2 - (5p^3+2q)x + (p^3-q) = 0$

5. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then, the value of r is
[IIT-JEE 2007]

(*a*) $\frac{2}{9}(p-q)(2q-p)$ (*b*) $\frac{2}{9}(q-p)(2p-q)$
(*c*) $\frac{2}{9}(q-2p)(2q-p)$ (*d*) $\frac{2}{9}(2p-q)(2q-p)$

6. If a, b, c are the sides of a triangle ABC such that $x^2 - 2(a+b+c)x + 3\lambda(ab+bc+ca) = 0$ has real roots, then
[IIT-JEE 2006]

(*a*) $\lambda < \frac{4}{3}$ (*b*) $\lambda > \frac{5}{3}$
(*c*) $\lambda \in \left(\frac{4}{3}, \frac{5}{3}\right)$ (*d*) $\lambda \in \left(\frac{1}{3}, \frac{5}{3}\right)$

7. If one root is square of the other root of the equation $x^2 + px + q = 0$, then the relation between p and q is **[IIT-JEE 2004]**

(*a*) $p^3 - q(3p-1) + q^2 = 0$ (*b*) $p^3 - q(3p+1) + q^2 = 0$
(*c*) $p^3 + q(3p-1) + q^2 = 0$ (*d*) $p^3 + q(3p+1) + q^2 = 0$

8. The set of all real numbers x for which $x^2 - |x+2| + x > 0$
[IIT-JEE 2002]

(*a*) $(-\infty, -2) \cup (2, \infty)$ (*b*) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
(*c*) $(-\infty, -1) \cup (1, \infty)$ (*d*) $(\sqrt{2}, \infty)$

9. The number of solutions of $\log_4(x-1) = \log_2(x-3)$ is
[IIT-JEE 2001]

(*a*) 3 (*b*) 1 (*c*) 2 (*d*) 0

10. For the equation $3x^2 + px + 3 = 0, p > 0$, if one of the root is square of the other, then p is equal to **[IIT-JEE 2000]**

(*a*) 1/3 (*b*) 1 (*c*) 3 (*d*) 2

11. If α and $\beta (\alpha < \beta)$ are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then **[IIT-JEE 2000]**

(*a*) $0 < \alpha < \beta$ (*b*) $\alpha < 0 < \beta < |\alpha|$
(*c*) $\alpha < \beta < 0$ (*d*) $\alpha < 0 < |\alpha| < \beta$

12. The equation $\sqrt{x+1} - \sqrt{x-1} = \sqrt{4x-1}$ has **[IIT-JEE 1997]**

(*a*) no solution (*b*) one solution
(*c*) two solutions (*d*) more than two solutions

13. The equation $x^{\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}$ has **[IIT-JEE 1989]**

(*a*) at least one real solution
(*b*) exactly three real solutions
(*c*) exactly one irrational solution
(*d*) complex roots

14. If α and β are the roots of $x^2+px+q=0$ and α^4,β^4 are the roots of $x^2-rx+s=0$, then the equation $x^2-4qx+2q^2-r=0$ has always **[IIT-JEE 1989]**

(*a*) two real roots

(*b*) two positive roots

(*c*) two negative roots

(*d*) one positive and one negative root

15. The equation $x-\frac{2}{x-1}=1-\frac{2}{x-1}$ has **[IIT-JEE 1984]**

(*a*) one solution (*b*) no solution

(*c*) two equal roots (*d*) infinitely many roots

16. For real *x*, the function $\frac{(x-a)(x-b)}{(x-c)}$ will assume all real values provided **[IIT-JEE 1984]**

(*a*) $a>b>c$ (*b*) $a<b<c$

(*c*) $a>c<b$ (*d*) $a\le c\le b$

17. The number of real solutions of the equation $|x|^2-3|x|+2=0$ is **[IIT-JEE 1981]**

(*a*) 4 (*b*) 1 (*c*) 3 (*d*) 2

18. Both the roots of the equation

$(x-b)(x-c)+(x-a)(x-c)+(x-a)(x-b)=0$ are always **[IIT-JEE 1980]**

(*a*) positive (*b*) negative

(*c*) Real (*d*) None of the above

19. Let $a>0, b>0$ and $c>0$. Then, both the roots of the equation $ax^2+bx+c=0$ **[IIT-JEE 1979]**

(*a*) are real and negative (*b*) have negative real parts

(*c*) have positive real parts (*d*) None of the above

Comprehension/Passage Based

Let *p*, *q* be integers and let α, β be the roots of the equation, $x^2-x-1=0$ where $\alpha\ne\beta$. For $n=0, 1, 2,.....$, let $a_n=p\alpha^n+q\beta^n$.

FACT: If *a* and *b* are rational numbers and $a+b\sqrt{5}=0$, then $a=0=b$.

20. $a_{12}=$ **C-28.59 W-67.14 UA-4.27 [JEE Adv. 2017]**

(*a*) $a_{11}+2a_{10}$ (*b*) $2a_{11}+a_{10}$

(*c*) $a_{11}-a_{10}$ (*d*) $a_{11}+a_{10}$

FACT: If *a* and *b* are rational numbers and $a+b\sqrt{5}=0$ then $a=0=b$.

21. If $a_4=28$ then $p+2q=$

C-28.59 W-67.14 UA-4.27 [JEE Adv. 2017]

(*a*) 14 (*b*) 7 (*c*) 21 (*d*) 12

Assertion Reason/Statement Based

22. Let *a*, *b*, *c*, *p*, *q* be the real numbers. Suppose α, β are the roots of the equation $x^2+2px+q=0$ and $\alpha,\frac{1}{\beta}$ are the roots of the equation $ax^2+2bx+c=0$, where $\beta^2\notin\{-1,0,1\}$.

Statement-I $(p^2-q)(b^2-ac)\ge 0$

Statement-II $b\ne pa$ or $c\ne qa$ **[IIT-JEE 2008]**

(*a*) **Statement-I** is true, Statement-II is also true; Statement-II is the correct explanation of Statement-I

(*b*) **Statement-I** is true, Statement-II is also true; Statement-II is not the correct explanation of Statement-I

(*c*) **Statement-I** is true; Statement-II is false

(*d*) **Statement-I** is false; Statement-II is true.

Fill in the Blanks

23. The sum of all the real roots of the equation $|x-2|^2+|x-2|-2=0$ is..... . **[IIT-JEE 1997]**

24. If the products of the roots of the equation $x^2-3kx+2e^{2\ln k}-1=0$ is 7, then the roots are real for *k* = ... **[IIT-JEE 1984]**

25. If $2+i\sqrt{3}$ is a root of the equation $x^2+px+q=0$, where *p* and *q* are real, then $(p,q)=(\ldots,\ldots)$. **[IIT-JEE 1982]**

26. The coefficient of x^{99} in the polynomial $(x-1)(x-2)\ldots(x-100)$ is.... **[IIT-JEE 1982]**

True/False

27. If $P(x)=ax^2+bx+c$ and $Q(x)=-ax^2+dx+c$, where $ac\ne 0$, then $P(x)\,Q(x)$ has atleast two real roots. **[IIT-JEE 1985]**

28. There exists a value of θ between 0 and 2π that satisfies the equation $\sin^4\theta-2\sin^2\theta+1=2$. **[IIT-JEE 1984]**

29. The equation $2x^2+3x+1=0$ has an irrational root. **[IIT-JEE 1983]**

Subjective

30. If $x^2-10ax-11b=0$ have roots *c* and *d*. $x^2-10cx-11d=0$ have roots *a* and *b*, then find $a+b+c+d$. **[IIT-JEE 2006]**

31. If $x^2+(a-b)x+(1-a-b)=0$ where $a,b\in R$, then find the values of *a* for which equation has unequal real roots for all values of *b*. **[IIT-JEE 2003]**

32. If α, β are the roots of $ax^2+bx+c=0,(a\ne 0)$ and α + δ, β + δ are the roots of $Ax^2+Bx+C=0,(A\ne 0)$ for some constant δ, then prove that **[IIT-JEE 2000]**

$$\frac{b^2-4ac}{a^2}=\frac{B^2-4AC}{A^2}$$

33. Let $f(x)=Ax^2+Bx+C$ where *A*, *B*, *C* are real numbers. Prove that if $f(x)$ is an integer whenever *x* is an integer, then the numbers 2*A*, *A* + *B* and *C* are all integers. Conversely, prove that if the numbers 2*A*, *A* + *B* and *C* are all integers, then $f(x)$ is an integer whenever *x* is an integer. **[IIT-JEE 1998]**

34. Find the set of all solutions of the equation $2^{|y|}-|2^{y-1}-1|=2^{y-1}+1$ **[IIT-JEE 1997]**

35. Find the set of all *x* for which $\frac{2x}{2x^2+5x+2}>\frac{1}{x+1}$ **[IIT-JEE 1987]**

36. Solve $|x^2+4x+3|+2x+5=0$ **[IIT-JEE 1987]**

37. Solve for $x:(5+2\sqrt{6})^{x^2-3}+(5-2\sqrt{6})^{x^2-3}=10$ **[IIT-JEE 1985]**

38. If one root of the quadratic equation $ax^2+bx+c=0$ is equal to the nth power of the other, then show that $(ac^n)^{\frac{1}{n+1}}+(a^nc)^{\frac{1}{n+1}}+b=0$ **[IIT-JEE 1983]**

39. If α and β are the roots of $x^2+px+q=0$ and γ, δ are the roots of $x^2+rx+s=0$, then evaluate $(\alpha-\gamma)(\beta-\gamma)(\alpha-\delta)(\beta-\delta)$ in terms of *p*, *q*, *r* and *s* **[IIT-JEE 1979]**

40. Solve $2\log_x a + \log_{ax} a + 3\log_b a = 0$, where $a > 0, b = a^2 x$

[IIT-JEE 1978]

41. If α and β are the roots of the equation $x^2 + px + 1 = 0; \gamma, \delta$ are the roots of $x^2 + qx + 1 = 0$, then $q^2 - p^2 = (\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta)$

[IIT-JEE 1978]

Condition for Common Roots

Single Correct

42. A value of b for which the equations $x^2 + bx - 1 = 0$, $x^2 + x + b = 0$ have one root in common is **[IIT-JEE 2011]**

(*a*) $-\sqrt{2}$ (*b*) $-i\sqrt{3}$ (*c*) $i\sqrt{5}$ (*d*) $\sqrt{2}$

Fill in the Blanks

43. If the quadratic equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ $(a \neq b)$ have a common root, then the numerical value of $a + b$ is... .

[IIT-JEE 1986]

True/False

44. If $x - r$ is a factor of the polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + ... + a_0$ repeated m times $(1 < m \leq n)$, then r is a root of $f'(x) = 0$ repeated m times. **[IIT-JEE 1983]**

Location of Roots

Single Correct

45. For all x, $x^2 + 2ax + (10 - 3a) > 0$ then the interval in which 'a' lies is **[IIT-JEE 2004]**

(*a*) $a < -5$ (*b*) $-5 < a < 2$

(*c*) $a > 5$ (*d*) $a < -2$

46. If $b > a$, then the equation $(x - a)(x - b) - 1 = 0$ has

[IIT-JEE 2000]

(*a*) both root in (a, b)

(*b*) both root in $(-\infty, a)$

(*c*) both root in $(b, +\infty)$

(*d*) one root in $(-\infty, a)$ and other root in (b, ∞)

47. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then **[IIT-JEE 1999]**

(*a*) $a < 2$ (*b*) $2 \leq a \leq 3$ (*c*) $3 < a \leq 4$ (*d*) $a > 4$

48. Let $f(x)$ be a quadratic expression which is positive for all real values of x. If $g(x) = f(x) + f'(x) + f''(x)$ then for any real x **[IIT-JEE 1990]**

(*a*) $g(x) < 0$ (*b*) $g(x) > 0$

(*c*) $g(x) = 0$ (*d*) $g(x) \geq 0$

Numerical/Integer Types

49. The smallest value of k for which both the roots of the equation $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is ..... **[IIT-JEE 2009]**

Subjective

50. If $x^2 + (a - b)x + (1 - a - b) = 0$ where $a, b \in R$, then find the values of a for which equation has unequal real roots for all values of b. **[IIT-JEE 2003]**

51. Let a, b, c be real. If $ax^2 + bx + c = 0$ has two real roots α and β, where $\alpha < -1$ and $\beta > 1$, then show that $1 + \frac{c}{a} + \left|\frac{b}{a}\right| < 0$

[IIT-JEE 1995]

52. Find all real values of x which satisfy $x^2 - 3x + 2 > 0$ and $x^2 - 2x - 4 \leq 0$ **[IIT-JEE 1983]**

Special Forms and Transformations of Roots

Single Correct

53. Let α, β be the roots of the equation, $(x - a)(x - b) = c, c \neq 0$. Then the roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are

[IIT-JEE 1992]

(*a*) a, c (*b*) b, c

(*c*) a, b (*d*) $a + c, b + c$

54. Let a, b, c be real numbers, $a \neq 0$. If α is a root of $a^2x^2 + bx + c = 0$, β is the root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then the equation $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies

[IIT-JEE 1989]

(*a*) $\gamma = \frac{\alpha + \beta}{2}$ (*b*) $\gamma = \alpha + \frac{\beta}{2}$

(*c*) $\alpha = \gamma$ (*d*) $\alpha < \gamma < \beta$

55. If $a + b + c = 0$, then the quadratic equation $3ax^2 + 2bx + c = 0$ has **[IIT-JEE 1983]**

(*a*) at least one root in (0, 1)

(*b*) one root in (2, 3)and the other in (–2, –1)

(*c*) imaginary roots

(*d*) None of the above

56. The largest interval for which $x^{12} - x^9 + x^4 - x + 1 > 0$ is **[IIT-JEE 1982]**

(*a*) $-4 < x \leq 0$ (*b*) $0 < x < 1$

(*c*) $-100 < x < 100$ (*d*) $-\infty < x < \infty$

Multiple Correct

57. Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following interval(s) is/are a subset of S ?

[JEE Adv. 2015]

(*a*) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (*b*) $\left(-\frac{1}{\sqrt{5}}, 0\right)$

(*c*) $\left(0, \frac{1}{\sqrt{5}}\right)$ (*d*) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

58. Let $a \in R$ and $f : R \to R$ be given by $f(x) = x^5 - 5x + a$.

Then, C-32.05 W-56.96 UA-10.99 **[JEE Adv. 2014]**

(*a*) $f(x)$ has three real roots, if $a > 4$

(*b*) $f(x)$ has only one real root, if $a > 4$

(*c*) $f(x)$ has three real roots, if $a < -4$

(*d*) $f(x)$ has three real roots, if $-4 < a < 4$

Comprehension/Passage Based

Passage 1

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of $f(x)$ and let $t = |s|$. **[IIT-JEE 2010]**

59. The real numbers s lies in the interval

(*a*) $\left(-\frac{1}{4},0\right)$ (*b*) $\left(-11,-\frac{3}{4}\right)$ (*c*) $\left(-\frac{3}{4},-\frac{1}{2}\right)$ (*d*) $\left(0,\frac{1}{4}\right)$

60. The area bounded by the curve $y = f(x)$ and the lines $x = 0$, $y = 0$ and $x = t$, lies in the interval **[IIT-JEE 2010]**

(*a*) $\left(\frac{3}{4},3\right)$ (*b*) $\left(\frac{21}{64},\frac{11}{16}\right)$

(*c*) (9, 10) (*d*) $\left(0,\frac{21}{64}\right)$

61. The function $f'(x)$ is **[IIT-JEE 2010]**

(*a*) increasing in $\left(-t,-\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4},t\right)$

(*b*) decreasing in $\left(-t,-\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4},t\right)$

(*c*) increasing in $(-t, t)$

(*d*) decreasing in $(-t, t)$

Passage 2

If a continuous function f defined on the real line R, assumes positive and negative values in R, then the equation $f(x) = 0$ has a root in R. For example, if it is known that a continuous function f on R is positive at some point and its minimum values is negative, then the equation $f(x) = 0$ has a root in R, Consider $f(x) = ke^x - x$ for all real x where k is real constant.

62. The line $y = x$ meets $y = ke^x$ for $k \le 0$ at **[IIT-JEE 2007]**

(*a*) no point (*b*) one point

(*c*) two points (*d*) more than two points

63. The positive value of k for which $ke^x - x = 0$ has only one root is **[IIT-JEE 2007]**

(*a*) $\frac{1}{e}$ (*b*) 1 (*c*) e (*d*) $\log_e 2$

64. For $k > 0$ the set of all values of k for which $ke^x - x = 0$ **[IIT-JEE 2007]**

(*a*) $\left(0,\frac{1}{e}\right)$ (*b*) $\left(\frac{1}{e},1\right)$ (*c*) $\left(\frac{1}{e},\infty\right)$ (*d*) (0, 1)

Numerical/Integer Types

65. For $x \in R$, the number of real roots of the equation

$3x^2 - 4|x^2 - 1| + x - 1 = 0$ is _____ .

C-38.81 W-55.58 UA-5.61 [JEE Adv. 2021]

True/False

66. If $a < b < c < d$, then the roots of the equation $(x - a)(x - c) + 2(x - b)(x - d) = 0$ are real and distinct.

[IIT-JEE 1984]

Subjective

67. Let a, b and c be real numbers with $a \neq 0$ and let α, β be the roots of the equation $ax^2 + bx + c = 0$. Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β **[IIT-JEE 2001]**

68. Let $-1 \le p \le 1$. Show that the equation $4x^3 - 3x - p = 0$ has a unique root in the interval [1/2, 1] and identify it. **[IIT-JEE 2001]**

ANSWER KEY

1. (*d*)	**2.** (*c*)	**3.** (*d*)	**4.** (*b*)	**5.** (*d*)	**6.** (*a*)	**7.** (*a*)	**8.** (*b*)	**9.** (*b*)	**10.** (*c*)
11. (*b*)	**12.** (*a*)	**13.** (*b*)	**14.** (*a*)	**15.** (*b*)	**16.** (*d*)	**17.** (*a*)	**18.** (*c*)	**19.** (*b*)	**20.** (*d*)
21. (*d*)	**22.** (*b*)	**23.** [4]	**24.** [2]	**25.** [–4, 7]	**26.** [–5050]	**27.** [True]	**28.** [False]	**29.** [False]	**42.** (*b*)
43. [–1]	**44.** [False]	**45.** (*b*)	**46.** (*d*)	**47.** (*a*)	**48.** (*b*)	**49.** [2]	**53.** (*c*)	**54.** (*d*)	**55.** (*a*)
56. (*d*)	**57.** (*a, d*)	**58.** (*b, d*)	**59.** (*c*)	**60.** (*a*)	**61.** (*b*)	**62.** (*b*)	**63.** (*a*)	**64.** (*a*)	**65.** [4]
66. [True]									

2 COMPLEX NUMBERS

JEE-Advanced

Definition Based

Single Correct

1. If $x+iy=\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix}$, find x and y. **[IIT-JEE 1998]**

(a) $x = 3, y = 1$ (b) $x = 1, y = 1$
(c) $x = 0, y = 3$ (d) $x = 0, y = 0$

2. The value of sum $\sum_{n=1}^{13}\left(i^n+i^{n+1}\right)$, where $i=\sqrt{-1}$, equals **[IIT-JEE 1998]**

(a) i (b) $i-1$
(c) $-i$ (d) 0

3. For positive integers n_1, n_2 the value of expression $(1+i)^{n_1}+(1+i^3)^{n_1}+(1+i^5)^{n_2}+(1+i^7)^{n_2}$, here $i=\sqrt{-1}$ is a real number, if and only if **[IIT-JEE 1996]**

(a) $n_1 = n_2 + 1$ (b) $n_1 = n_2 - 1$
(c) $n_1 = n_2$ (d) $n_1 > 0, n_2 > 0$

4. The smallest positive integer n for which $\left(\frac{1+i}{1-i}\right)^n=1$, is **[IIT-JEE 1980]**

(a) 8 (b) 16
(c) 4 (d) None of these

Numerical/Integer Types

5. Let $A=\left\{\frac{1967+1686i\sin\theta}{7-3i\cos\theta}:\theta\in\mathbb{R}\right\}$. If A contains exactly one positive integer n, then the value of n is

C-10.77 W-57.72 UA-31.51 **[JEE Adv. 2023]**

Properties of Conjugate and Modulus

Single Correct

6. Let complex numbers α and $1/\bar{\alpha}$ lies on circles $(x-x_0)^2+(y-y_0)^2=r^2$ and $(x-x_0)^2+(y-y_0)^2=4r^2$, respectively. If $z_0=x_0+iy_0$ satisfies the equation $2|z_0|^2=r^2+2$ then $|\alpha|$ is equal to

C-26.76 W 63.3 UA-9.94 **[JEE Adv. 2013]**

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$
(c) $\frac{1}{\sqrt{7}}$ (d) $\frac{1}{3}$

7. Let z be a complex number such that the imaginary part of z is non-zero and $a = z^2 + z + 1$ is real. Then, a cannot take the value

C-14.09 W-45.71 UA-40.19 **[IIT-JEE 2012]**

(a) -1 (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

8. Let $z = x + iy$ be a complex number where, x and y are integers. Then, the area of the rectangle whose vertices are the root of the equation $z\bar{z}^3+\bar{z}z^3=350$, is **[IIT-JEE 2009]**

(a) 48 (b) 32
(c) 40 (d) 80

9. If $w = \alpha + i\beta$, where $\beta \neq 0$ and $z \neq 1$, satisfies the condition that $\left(\frac{w-\bar{w}z}{1-z}\right)$ is purely real, then the set of values of z is **[IIT-JEE 2006]**

(a) $|z| = 1, z \neq 2$ (b) $|z| = 1$ and $z \neq 1$
(c) $z = \bar{z}$ (d) None of these

10. If $|z| = 1$ and $w=\frac{z-1}{z+1}$ (where, $z \neq -1$), then $Re(w)$ is **[IIT-JEE 2003]**

(a) 0 (b) $\frac{1}{|z+1|^2}$
(c) $\left|\frac{1}{z+1}\right|\cdot\frac{1}{|z+1|^2}$ (d) $\frac{\sqrt{2}}{|z+1|^2}$

11. If z_1, z_2 and z_3 are complex numbers such that $|z_1|=|z_2|=|z_3|=\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right|=1$, then $|z_1+z_2+z_3|$ is **[IIT-JEE 2000]**

(a) equal to 1 (b) less than 1
(c) greater than 3 (d) equal to 3

12. The complex numbers numbers $\sin x + i\cos 2x$ and $\cos x - i\sin 2x$ are conjugate to each other, for **[IIT-JEE 1988]**

(a) $x = n\pi$ (b) $x = 0$
(c) $x = (n + 1/2)\pi$ (d) no value of x

13. If $z = x + iy$ and $w = (1 - iz)/(z - i)$, then $|w| = 1$ implies that, in the complex plane **[IIT-JEE 1989]**

(*a*) z lies on the imaginary axis

(*b*) z lies on the real axis

(*c*) z lies on the unit circle

(*d*) None of these

Multiple Correct

14. Lei $\bar{z}$ denote the complex conjugate of a complex number z. If z is a non-zero complex number for which both real and imaginary parts of $(\bar{z})^2 + \frac{1}{z^2}$ are integers, then which of the following is/are possible value(s) of $|z|$? C-2.14 W-26.93 UA-70.92 **[JEE Adv. 2022]**

(*a*) $\left(\frac{43+3\sqrt{205}}{2}\right)^{\frac{1}{4}}$ (*b*) $\left(\frac{7+\sqrt{33}}{4}\right)^{\frac{1}{4}}$

(*c*) $\left(\frac{9+\sqrt{65}}{4}\right)^{\frac{1}{4}}$ (*d*) $\left(\frac{7+\sqrt{13}}{6}\right)^{\frac{1}{4}}$

15. Let S be the set of all complex numbers z satisfying $|z^2 + z + 1| = 1$. Then which of the following statements is/are true?

C-8.01 W-31.31 UA-49.59 PC-11.09 **[JEE Adv. 2020]**

(*a*) $\left|z+\frac{1}{2}\right| \le \frac{1}{2}$ fall all $z \in S$

(*b*) $|z| \le 2$ for all $z \in S$

(*c*) $\left|z+\frac{1}{2}\right| \ge \frac{1}{2}$ fall all $z \in S$

(*d*) The set S has exactly four elements

16. Let s,t,r be non-zero complex numbers and L be the set of solutions $z = x + iy\left(x, y \in R, i = \sqrt{-1}\right)$ of the equation $sz + t\bar{z} + r = 0$, where $\bar{z} = x - iy$. Then, which of the following statement(s) is (are) TRUE?

C-1.18 W-27.03 UA-56.41 PC-15.38 **[JEE Adv. 2018]**

(*a*) If L has exactly one element, then $|s| \neq |t|$

(*b*) If $|s| = |t|$, then L has infinitely many elements

(*c*) The number of elements in $L \cap \{z : |z - 1 + i| = 5\}$ is at most 2

(*d*) If L has more than one element, then L has infinitely many elements

17. Let a,b,x and y be real numbers such that $a - b = 1$ and $y \neq 0$. If the complex number $z = x + iy$ satisfies $Im\left(\frac{az+b}{z+1}\right) = y$, then which of the following is(are) possible value(s) of x ?

C-31.5 W-13.18 UA-49.89 **[JEE Adv. 2017]**

(*a*) $1-\sqrt{1+y^2}$ (*b*) $-1-\sqrt{1-y^2}$

(*c*) $1+\sqrt{1+y^2}$ (*d*) $-1+\sqrt{1-y^2}$

18. Let z_1 and z_2 be complex numbers such that $z_1 \neq z_2$ and $|z_1| = |z_2|$. If z_1 has positive real part and z_2 has negative imaginary part, then $\frac{z_1+z_2}{z_1-z_2}$ may be **[IIT-JEE 1986]**

(*a*) zero (*b*) real and positive

(*c*) real and negative (*d*) purely imaginary

19. If $z_1 = a + ib$ and $z_2 = c + id$ are complex numbers such that $|z_1| = |z_2| = 1$ and $Re(z_1\bar{z}_2) = 0$, then the pair of complex numbers $w_1 = a + ic$ and $w_2 = b + id$ satisfies **[IIT-JEE 1985]**

(*a*) $|w_1| = 1$ (*b*) $|w_2| = 1$

(*c*) $Re(w_1\bar{w}_2) = 0$ (*d*) None of these

Match the Column

20. Let z be complex number satisfying $|z|^3 + 2z^2 + 4\bar{z} - 8 = 0$ where $\bar{z}$ denotes the complex conjugate of z. Let the imaginary part of z be non zero. C-13.5 W-32.06 UA-54.44 **[JEE Adv. 2023]**

Match List-I with List-II:

	List-I		List-I
A.	$\lvert z\rvert^2$ is equal to	I.	12
B.	$\lvert z-\bar{z}\rvert^2$ is equal to	II.	4
C.	$\lvert z\rvert^2 + \lvert z+\bar{z}\rvert^2$ is equal to	III.	8
D.	$\lvert z+1\rvert^2$ is equal to	IV.	10
		V.	7

Choose the correct answer from the options given below:

(*a*) A → I; B → III; C → V; D → IV

(*b*) A → II; B → I; C → III; D → V

(*c*) A → II; B → IV; C → V; D → I

(*d*) A → II; B → III; C → V; D → IV

21. Match the statements of Column-I with those of Column-II. Here, z takes values in the complex plane and *Im* (z) and *Re* (z) denote respectively, the imaging part and real part of $z\,i$ **[IIT-JEE 2010]**

	Column-I		Column-II
(A)	The set of points z satisfying $\lvert z - i\lvert z\rvert\rvert = \lvert z + i\lvert z\rvert\rvert$ is contained in or equal to	(p)	an ellipse with eccentricity 4/5
(B)	The set of points z satisfying $\lvert z+4\rvert+\lvert z-4\rvert=10$ is contained in or equal to	(q)	the set of points z satisfying $Im\, z = 0$
(C)	If $\lvert w\rvert = 2$, then the set of points $z = w - 1/w$ is contained in or equal to	(r)	the set of points z satisfying $\lvert Im z\rvert \le 1$
(D)	If $\lvert w\rvert = 1$, then the set of points $z = w + 1/w$ is contained in or equal to	(s)	the set of points z satisfying $\lvert Re z\rvert < 2$
		(t)	the set of point z satisfying $\lvert z\rvert \le 3$

Numerical/Integer Types

22. Let z be a complex number with non-zero imaginary part. If $\frac{2+3z+4z^2}{2-3z+4z^2}$ is a real number, then the value of $|z|^2$ is_____.

C-7.05 W-56.77 UA-36.18 **[JEE Adv. 2022]**

23. Let z denote the complex conjugate of a complex number z and let $i = \sqrt{-1}$ in the set of complex numbers, the number of distinct roots of the equation $\bar{z} - z^2 = i(\bar{z} + z^2)$ is_____

C-10.03 W-59.83 UA-30.14 **[JEE Adv. 2022]**

24. If z is any complex number satisfying $|z - 3 - 2i| \le 2$, then the minimum value of $|2z - 6 + 5i|$ is ..... **[IIT-JEE 2011]**

Fill in the Blanks

25. If α,β,γ are the cube roots of p, $p<0$, then for any x,y and z then $\frac{x\alpha+y\beta+z\gamma}{x\beta+y\gamma+z\alpha}=\ldots$ **[IIT-JEE 1990]**

26. For any two complex numbers z_1, z_2 and any real numbers a and b, $|az_1-bz_2|^2+|bz_1+az_2|^2=\ldots$ **[IIT-JEE 1988]**

27. If the expression $\frac{\left[\sin\left(\frac{x}{2}\right)+\cos\left(\frac{x}{2}\right)-i\tan\left(\frac{x}{2}\right)\right]}{\left[1+2i\sin\left(\frac{x}{2}\right)\right]}$ is real, then the set of all possible values of x is .... **[IIT-JEE 1978]**

True/False

28. For complex numbers $z_1=x_1+iy_1$ and $z_2=x_2+iy_2$ we write $z_1\cap z_2$, if $x_1\le x_2$ and $y_1\le y_2$. Then, for all complex numbers z with $1\cap z$, we have $\frac{1-z}{1+z}\cap 0$. **[IIT-JEE 1981]**

Subjective

29. Prove that there exists no complex number z such that $|z|<1/3$ and $\sum_{r=1}^{n}a_rz^r=1$, where $|a_r|<2$. **[IIT-JEE 2003]**

30. If z_1 and z_2 are two complex numbers such that $|z_1|<1<|z_2|$, then prove that $\left|\frac{1-z_1\bar{z}_2}{z_1-z_2}\right|<1$ **[IIT-JEE 2003]**

31. For complex numbers z and w, prove that $|z|^2w-|w|^2z=z-w$, if and only if $z=w$ or $z\bar{w}=1$. **[IIT-JEE 1999]**

32. Find all non-zero complex numbers z satisfying $\bar{z}=iz^2$. **[IIT-JEE 1996]**

33. If $iz^3+z^2-z+i=0$, then show that $|z|=1$. **[IIT-JEE 1995]**

34. A relation R on the set of complex numbers is defined by z_1Rz_2, if and only if $\frac{z_1-z_2}{z_1+z_2}$ is real. Show that R is an equivalence relation. **[IIT-JEE 1982]**

35. Find the real values of x and y for which the following equation is satisfied $\frac{(1+i)x-2i}{3+i}+\frac{(2-3i)y+i}{3-i}=i$. **[IIT-JEE 1980]**

36. Express $\frac{1}{(1-\cos\theta)+2i\sin\theta}$ in the form A + iB. **[IIT-JEE 1979]**

37. If $x+iy=\sqrt{\frac{a+ib}{c+id}}$, prove that $(x^2+y^2)^2=\frac{a^2+b^2}{c^2+d^2}$ **[IIT-JEE 1978]**

Properties of Amplitude

Single Correct

38. Let z and w be two non-zero complex numbers such that $|z|=|w|$ and $\arg(z)+\arg(w)=\pi$, then z equals **[IIT-JEE 1995]**

(*a*) w (*b*) $-w$

(*c*) $\bar{w}$ (*d*) $-\bar{w}$

39. If z_1 and z_2 are two non-zero complex numbers such that $|z_1+z_2|=|z_1|+|z_2|$, then $\arg(z_1)-\arg(z_2)$ is equal to **[IIT-JEE 1987]**

(*a*) $-\pi$ (*b*) $-\pi/2$

(*c*) 0 (*d*) $\pi/2$

Match the Column

40. Match the conditions/expressions in Column-I with statement in Column-II ($z\ne 0$ is a complex number) **[IIT-JEE 1992]**

Column-I		Column-II	
A.	$Re(z)=0$	*p.*	$Re(z^2)=0$
B.	$\arg(z)=\pi/4$	*q.*	$Im(z^2)=0$
		r.	$Re(z^2)=Im(z^2)$

Subjective

41. $|z|\le 1$, $|w|\le 1$, then show that $|z-w|^2\le(|z|-|w|)^2+(\arg z-\arg w)^2$ **[IIT-JEE 1995]**

Complex Number-Geometry

Single Correct

42. Let $\theta_1,\theta_2,\ldots,\theta_{10}$ be positive valued angles (in radian) such that $\theta_1+\theta_2+\ldots+\theta_{10}=2\pi$.
Define the complex numbers $z_1=e^{i\theta_1}, z_k=z_{k-1}e^{i\theta_k}$ for $k=2,3,\ldots,10$, where $i=\sqrt{-1}$. Consider the statements P and Q given below:

$$P:|z_2-z_1|+|z_3-z_2|+\ldots+|z_{10}-z_9|+|z_1-z_{10}|\le 2\pi$$

$$Q:|z_2^2-z_1^2|+|z_3^2-z_2^2|+\ldots+|z_{10}^2-z_9^2|+|z_1^2-z_{10}^2|\le 4\pi$$

Then, C-16.06 W-18.26 UA-65.69 **[JEE Adv. 2021]**

(*a*) P is **TRUE** and Q is **FALSE**

(*b*) Q is **TRUE** and P is **FALSE**

(*c*) Both P and Q are **TRUE**

(*d*) Both P and Q are **FALSE**

43. Let S be the set of all complex numbers z satisfying $|z-2+i|\ge\sqrt{5}$. If the complex number z_0 is such that $\frac{1}{|z_0-1|}$ is the maximum of the set $\left\{\frac{1}{|z-1|}:z\in S\right\}$ then the principal argument of $\frac{4-z_0-\bar{z}_0}{z_0-\bar{z}_0+2i}$ is

C-15.22 W-26.59 UA-58.19 **[JEE Adv. 2019]**

(*a*) $\pi/4$ (*b*) $3\pi/4$

(*c*) $-\pi/2$ (*d*) $\pi/2$

44. For all complex numbers z_1,z_2 satisfying $|z_1|=12$ and $|z_2-3-4i|=5$, the minimum value of $|z_1-z_2|$ is **[IIT-JEE 2002]**

(*a*) 0 (*b*) 2

(*c*) 7 (*d*) 17

45. Let z and w be two complex numbers such that $|z|\le 1$, $|w|\le 1$ and $|z+iw|=|z-i\bar{w}|=2$, then z equals **[IIT-JEE 1995]**

(*a*) 1 or i (*b*) i or $-i$

(*c*) 1 or -1 (*d*) i or -1

46. If a,b,c and u,v,w are the complex numbers representing the vertices of two triangles such that $c=(1-r)a+rb$ and $w=(1-r)u+rv$, where r is a complex number, then the two triangles **[IIT-JEE 1985]**

(*a*) have the same area (*b*) are similar

(*c*) are congruent (*d*) None of these

47. The points z_1, z_2, z_3 and z_4 in the complex plane are the vertices of a parallelogram taken in order, if and only if **[IIT-JEE 1983]**

(*a*) $z_1 + z_4 = z_2 + z_3$ (*b*) $z_1 + z_3 = z_2 + z_4$

(*c*) $z_1 + z_2 = z_3 + z_4$ (*d*) None of these

48. The inequality $|z - 4| < |z - 2|$ represents the region given by **[IIT-JEE 1982]**

(*a*) $Re(z) \geq 0$ (*b*) $Re(z) < 0$

(*c*) $Re(z) > 0$ (*d*) None of these

49. The complex numbers $z = x + iy$ which satisfy the equation $\left|\frac{z-5i}{z+5i}\right| = 1$, lie on **[IIT-JEE 1981]**

(*a*) the *x*-axis

(*b*) the straight line $y = 5$

(*c*) a circle passing through the origin

(*d*) None of the above

Multiple Correct

50. For any complex number $w = c + id$, let $\arg(w) \in (-\pi, \pi]$, where $i = \sqrt{-1}$. Let α and β be real numbers such that for all complex numbers $z = x + iy$ satisfying $\arg\left(\frac{z+\alpha}{z+\beta}\right) = \frac{\pi}{4}$, the ordered pair (x, y) lies on the circle

$$x^2 + y^2 + 5x - 3y + 4 = 0$$

Then which of the following statements is (are) TRUE?

C-12.75 W-24.93 UA-55.54 PC-6.79 [JEE Adv. 2021]

(*a*) $\alpha = -1$ (*b*) $\alpha\beta = 4$ (*c*) $\alpha\beta = -4$ (*d*) $\beta = 4$

51. For a non-zero complex number z, let *arg*(z) denote the principal argument with $-\pi < arg(z) \leq \pi$. Then, which of the following statement(s) is/are false?

C-1.83 W-28.69 UA-35.93 PC-33.55 [JEE Adv. 2018]

(*a*) $arg(-1 - i) = \pi/4$, where $i = \sqrt{-1}$

(*b*) The function $f: R \to (-\pi, \pi]$, defined by $f(t) = arg(-1 + it)$ for all $t \in R$, is continuous at all points of R, where $i = \sqrt{-1}$

(*c*) For any two non-zero complex numbers z_1 and z_2 $arg\left(\frac{z_1}{z_2}\right) - arg(z_1) + arg(z_2)$ is an integer multiple of 2π.

(*d*) For any three given distinct complex numbers z_1, z_2 and z_3, the locus of the point z satisfying the condition $\left(\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}\right) = \pi$, lies on a straight line.

52. Let $a, b \in R$ and $a^2 + b^2 \neq 0$. Suppose

$S = \left\{z \in C : z = \frac{1}{a+ibt}, t \in R, t \neq 0\right\}$, where $i = \sqrt{-1}$. If $z = x + iy$ and $z \in S$, then (x, y) lies on

C-5.79 W-11.34 UA-56.29 PC-26.57 [JEE Adv. 2016]

(*a*) the circle with radius $\frac{1}{2a}$ and center $\left(\frac{1}{2a}, 0\right)$ for $a > 0, b \neq 0$

(*b*) the circle with radius $-\frac{1}{2a}$ and center $\left(-\frac{1}{2a}, 0\right)$ for $a < 0, b \neq 0$

(*c*) the *x*-axis for $a \neq 0, b = 0$

(*d*) the *y*-axis for $a = 0, b = 0$

53. Let z_1 and z^2 be two distinct complex numbers and let $z = (1 - t)z_1 + tz_2$ for some real number t with $0 < t < 1$. If $arg(w)$ denotes the principal argument of a non-zero complex number w, then **[IIT-JEE 2010]**

(*a*) $|z - z_1| + |z - z_2| = |z_1 - z_2|$

(*b*) $arg\,|z - z_1| = arg\,|z - z_2|$

(*c*) $\begin{vmatrix} z - z_1 & \bar{z} - \bar{z}_1 \\ z_2 - z_1 & \bar{z}_2 - z_1 \end{vmatrix} = 0$

(*d*) $arg\,|z - z_1| = arg\,|z_2 - z_1|$

Comprehension/Passage Based

Let $S = S_1 \cap S_2 \cap S_3$, where

$$S_1 = \{z \in \mathbb{C} : |z| < 4\}, S_2 = \left\{z \in \mathbb{C} : \text{Im}\left[\frac{z - 1 + \sqrt{3}i}{1 - \sqrt{3}i}\right] > 0\right\}$$

and $S_3 = \{z \in \mathbb{C} : \text{Re}z > 0\}$.

54. $\min_{z \in S}|1 - 3i - z|$ is equal to

C-13.83 W-22.76 UA-63.41 [JEE Adv. 2013]

(*a*) $\frac{2-\sqrt{3}}{2}$ (*b*) $\frac{2+\sqrt{3}}{2}$

(*c*) $\frac{3-\sqrt{3}}{2}$ (*d*) $\frac{3+\sqrt{3}}{2}$

55. Area of S is equal to

C-15.14 W-28.63 UA-56.23 [JEE Adv. 2013]

(*a*) $\frac{10\pi}{3}$ (*b*) $\frac{20\pi}{3}$

(*c*) $\frac{16\pi}{3}$ (*d*) $\frac{32\pi}{3}$

Numerical/Integer Types

56. For a complex number z, let *Re*(z) denote the real part of z. Let S be the set of all complex numbers z satisfying $z^4 - |z|^4 = 4iz^2$, where $i = \sqrt{-1}$. Then the minimum possible value of $|z_1 - z_2|^2$, where $z_1, z_2 \in S$ with $Re(z_1) > 0$ and $Re(z_2) < 0$ is _____.

C-3.49 W-16.03 UA-80.48 [JEE Adv. 2020]

True/False

57. If three complex numbers are in AP. Then, they lie on a circle in the complex plane **[IIT-JEE 1985]**

58. Equilateral triangle such that $|z_1| = |z_2| = |z_3|$, then $z_1 + z_2 + z_3 = 0$. **[IIT-JEE 1984]**

Subjective

59. Find the center and radius of the circle formed by all the points represented by $z = x + iy$ satisfying the relation $\left|\frac{z-\alpha}{z-\beta}\right| = k\,(k \neq 1)$, where α and β are the constant complex numbers given by $\alpha = \alpha_1 + i\alpha_2$, $\beta = \beta_1 + i\beta_2$. **[IIT-JEE 2004]**

60. Let $z_1 = 10 + 6i$ and $z_2 = 4 + 6i$. If z is any complex number such that the argument of $(z - z_1)/(z - z_2)$ is $\pi/4$, then prove that $|z - 7 - 9i| = 3\sqrt{2}$. **[IIT-JEE 1991]**

Rotation Theorem

Single Correct

61. A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point z_1. From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{i} + \hat{j}$ and then it moves through an angle $\frac{\pi}{2}$ in anti-clockwise direction on a circle with center at origin, to reach a point z_2. The point z_2 is given by **[IIT-JEE 2008]**

(*a*) $6 + 7i$ (*b*) $-7 + 6i$

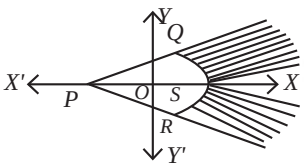
(*c*) $7 + 6i$ (*d*) $-6 + 7i$

62. A man walks a distance of 3 units from the origin towards the North-East (N45°E) direction. From there, he walks a distance of 4 units towards the North-West (N45°W) direction to reach a point P. Then, the position of P in the Argand plane **[IIT-JEE 2007]**

(*a*) $3e^{i\pi/4} + 4i$ (*b*) $(3-4i)e^{\pi/4}$

(*c*) $(4+3i)e^{j\pi/4}$ (*d*) $(3+4i)e^{i\pi/4}$

63. The shaded region, where $P = (-1,0), Q = \left(-1+\sqrt{2}, \sqrt{2}\right)$ $R = \left(-1+\sqrt{2}, -\sqrt{2}\right), S(1,0)$ is represented by **[IIT-JEE 2005]**



(*a*) $|z+1| > 2, |arg(z+1)| < \frac{\pi}{4}$

(*b*) $|z+1| < 2,\ \arg(z+1) < \frac{\pi}{2}$

(*c*) $|z+1| > 2, |arg(z+1)| > \frac{\pi}{4}$

(*d*) $|z-1| > 2, |arg(z+1)| > \frac{\pi}{4}$

64. If $0 < \alpha < \frac{\pi}{2}$ is a fixed angle. If $P = (\cos\theta, \sin\theta)$ and $Q = \left\{cos\left(\alpha \frac{2}{\theta}\right), sin(\alpha - \theta)\right\}$, then Q is obtained from P by **[IIT-JEE 2002]**

(*a*) clockwise rotation around origin through an angle α

(*b*) anti-clockwise rotation around origin through an angle α

(*c*) reflection in the line through origin with slope $\tan\alpha$

(*d*) reflection in the line through origin with slope $tan\frac{\alpha}{2}$

65. The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ are the vertices of a triangle which is **[IIT-JEE 2001]**

(*a*) of area zero (*b*) right angled isosceles

(*c*) equilateral (*d*) obtuse angled isosceles

Fill in the Blanks

66. Suppose z_1, z_2, z_3 are the vertices of an equilateral triangle inscribed in the circle $|z| = 2$. If $z_1 = 1 + i\sqrt{3}$, then $z_2 = ...$, $z_3 = ...$ **[IIT-JEE 1994]**

67. $ABCD$ is a rhombus. Its diagonals AC and BD intersect at the point M and satisfy $BD = 2AC$. If the points D and M represent the complex numbers $1 + i$ and $2 - i$ respectively, then A represents the complex number ...or... **[IIT-JEE 1993]**

68. If a and b are real numbers between 0 and 1 such that the points $z_1 = a + i$, $z_2 = 1 + bi$ and $z_3 = 0$ form an equilateral triangle, then $a = ...$ and $b = ...$ **[IIT-JEE 1990]**

Subjective

69. If one of the vertices of the square circumscribing the circle $|z-1| = \sqrt{2}$ is $2 + \sqrt{3}$. Find the other vertices of the square **[IIT-JEE 2005]**

70. Let $\bar{b}z + b\bar{z} = c$, $b \neq 0$, be a line in the complex plane, where $\bar{b}$ is the complex conjugate of b. If a point z_1 is the reflection of the point z_2 through the line, then show that $c = \bar{z}_1 b + z_2 \bar{b}$. **[IIT-JEE 1997]**

71. Let z_1 and z_2 be the roots of the equation $z^2 + pz + q = 0$, where the coefficients p and q may be complex numbers. Let A and B represent z_1 and z_2 in the complex plane. If $\angle AOB = \alpha \neq 0$ and $OA = OB$, where O is the origin prove that $p^2 = 4q\cos^2\left(\frac{\alpha}{2}\right)$. **[IIT-JEE 1997]**

72. Complex numbers z_1, z_2, z_3 are the vertices A, B, C respectively of an isosceles right angled triangle with right angle at C. Show that $(z_1 - z_2)^2 = 2(z_1 - z_3)(z_3 - z_2)$ **[IIT-JEE 1986]**

73. Show that the area of the triangle on the argand diagram formed by the complex number z, iz and $z + iz$ is $\frac{1}{2}|z|^2$. **[IIT-JEE 1986]**

74. Prove that the complex numbers z_1, z_2 and the origin form an equilateral triangle only if $z_1^2 + z_2^2 - z_1 x_2 = 0$. **[IIT-JEE 1983]**

75. Let the complex numbers z_1, z_2 and z_3 be the vertices of an equilateral triangle. Let z_0 be the circumcentre of the triangle. Then, prove that $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$. **[IIT-JEE 1981]**

Demoivre's Theorem, Cube Root of Unity, Nth Roots of Unity

Single Correct

76. Let $z = \cos\theta + i\sin\theta$. Then, the value of $\sum_{m=1}^{15} Im\left(z^{2m-1}\right)$ at $\theta = 2°$ is **[IIT-JEE 2009]**

(*a*) $\frac{1}{sin2^\circ}$ (*b*) $\frac{1}{3sin2^\circ}$

(*c*) $\frac{1}{2sin2^a}$ (*d*) $\frac{1}{4sin2^\circ}$

77. If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on **[IIT-JEE 2007]**

(*a*) a line not passing through the origin

(*b*) $|z| = \sqrt{2}$

(*c*) the X-axis

(*d*) the Y-axis

78. If $\omega(\neq 1)$ be a cube root of unity and $(1+\omega^2)^n = (1+\omega^4)^n$, then the least positive value of n is **[IIT-JEE 2004]**

(*a*) 2 (*b*) 3

(*c*) 5 (*d*) 6

79. The minimum value of $|a + b\omega + c\omega^2|$, where a,b and c are all not equal integers and ω $(\neq 1)$(is a cube root of unity), is

[IIT-JEE 2005]

(*a*) $\sqrt{3}$ (*b*) $\frac{1}{2}$

(*c*) 1 (*d*) 0

80. Let z_1 and z_2 be n^{th} roots of unity which subtend a right angled at the origin, then n must be of the form (where, k is an integer)

[IIT-JEE 2001]

(*a*) $4k + 1$ (*b*) $4k + 2$

(*c*) $4k + 3$ (*d*) $4k$

81. If $i = \sqrt{-1}$, then $4+5\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{334}+3\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{365}$ is equal to

[IIT-JEE 1999]

(*a*) $1 - i\sqrt{3}$ (*b*) $-1 + i\sqrt{3}$

(*c*) $i\sqrt{3}$ (*d*) $-i\sqrt{3}$

82. If ω is an imaginary cube root of unity, then $(1 + \omega - \omega^2)^7$ is equal to **[IIT-JEE 1998]**

(*a*) 128ω (*b*) -128ω

(*c*) $128\omega^2$ (*d*) $-128\omega^2$

83. The value of $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7} - i\cos\frac{2\pi k}{7}\right)$ is **[IIT-JEE 1998]**

(*a*) -1 (*b*) 0

(*c*) $-i$ (*d*) i

84. If $\omega(\neq 1)$ is a cube root of unity and $(1 + \omega)^7 = A + B\omega$, then A and B are respectively **[IIT-JEE 1995]**

(*a*) 0, 1 (*b*) 1, 1

(*c*) 1, 0 (*d*) -1, 1

85. If $z = \left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5 + \left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5$, then **[IIT-JEE 1982]**

(*a*) $Re(z) = 0$ (*b*) $Im(z) = 0$

(*c*) $Re(z) > 0, Im(z) > 0$ (*d*) $Re(z) > 0, Im(z) < 0$

86. Let $\omega = \frac{1}{2} + i\frac{\sqrt{3}}{2}$, then value of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega \end{vmatrix}$ is **[IIT-JEE 1982]**

(*a*) 3ω (*b*) $3\omega(\omega - 1)$

(*c*) $3\omega^2$ (*d*) $3\omega(1 - \omega)$

Multiple Correct

87. Let $W = \frac{\sqrt{3}+i}{2}$ and $P = \{W^n : n = 1,2,3,\ldots\}$. Further $H_1 = \left\{z \in C : Re(z) > \frac{1}{2}\right\}$ and $H_2 = \left[z \in C : Re(z) < \frac{-1}{2}\right]$, where C is the set of all complex numbers. If $z_1 \in P \cap H_1, z_2 \in P \cap H_2$ and O represents the origin, then $\angle z_1Oz_2$ is equal to

C-4.63 W-28.5 UA-66.87 [JEE Adv. 2013]

(*a*) $\frac{\pi}{2}$ (*b*) $\frac{\pi}{6}$

(*c*) $\frac{2\pi}{3}$ (*d*) $\frac{5\pi}{6}$

Match the Column

88. Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i\sin\left(\frac{2k\pi}{10}\right)$; $k = 1, 2,\ldots.. 9$. **[IIT-JEE 2011]**

	Column-I	Column-II	
P.	For each z_k, there exists a z_j such that $z_k.z_j = 1$	(*i*)	True
Q.	There exists a $k \in \{1,2, \ldots., 9\}$ $z_1.z = z_k$ has no solution z in the set of complex numbers	(*ii*)	False
R.	$\frac{\lvert 1-z_1\rvert\lvert 1-z_2\rvert\ldots\lvert 1-z_9\rvert}{10}$ equal	(*iii*)	1
S.	$1-\sum_{k=1}^{9}\cos\left(\frac{2k\pi}{10}\right)$	(*iv*)	2

	P	Q	R	S
(*a*)	(*i*)	(*ii*)	(*iv*)	(*iii*)
(*b*)	(*ii*)	(*i*)	(*iii*)	(*iv*)
(*c*)	(*i*)	(*ii*)	(*iii*)	(*iv*)
(*d*)	(*ii*)	(*i*)	(*iv*)	(*iii*)

Numerical/Integer Types

89. Let $\omega \neq 1$ be a cube root of unity. Then the minimum of the set $\{|a + b\omega + c\omega^2|^2 : a, b, c$ distinct non-zero integers$\}$equals _____. **C-10.61 W-67.99 UA-21.4 [JEE Adv. 2019]**

90. For any integer k, let $\alpha_k = cos\left(\frac{k\pi}{7}\right) + isin\left(\frac{k\pi}{7}\right)$, where $i = \sqrt{-1}$.

The value of the expression

$$\frac{\sum_{k=1}^{12}|\alpha_{k+1} - \alpha_k|}{\sum_{k=1}^{3}|\alpha_{4k-1} - \alpha_{4k-2}|}$$ **[JEE Adv. 2015]**

91. Let $A_1, A_2, A_3\ldots..,A_8$ be the vertices of a regular octagon that lie on a circle of radius 2 . Let P be a point on the circle and let PA_i denote the distance between the points P and A_i for i = i = 1, 2...,8. If P varies over the circle, then the maximum value of the product $PA_1 . PA_2 \ldots.PA_8$, is **C-1.18 W-68.93 UA-29.89 [JEE Adv. 2023]**

92. Let ω be the complex numbers $\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$. Then the number of distinct complex number z satisfying $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$ is equal to _____. **[IIT-JEE 2010]**

Fill in the Blanks

93. The value of the expression $1(2-\omega)(2-\omega^2)+2(3-\omega)(3-\omega^2)+....+(n-1).(n-\omega)(n-\omega^2)$, where, ω is an imaginary cube root of unity, is _____. **[IIT-JEE 1996]**

True/False

94. The cube roots of unity where represented on Argand diagram form the vertices of an equilateral triangle. **[IIT-JEE 1988]**

Subjective

95. Let $\omega = e^{i\pi/3}$ and a, b, c, x, y, z be non-zero complex numbers such that $a+b+c=x$, $a+b\omega+c\omega^2=y$, $a+b\omega^2+c\omega=z$.

Then, the value of $\dfrac{|x|^2+|y|^2+|z|^2}{|a|^2+|b|^2+|c|^2}$ is..... **[IIT-JEE 2011]**

96. Let a complex number α, $\alpha \neq 1$, be a root of the equation $z^{p+q}-z^p-z^q+1=0$ where, p and q are distinct primes. Show that either $1+\alpha+\alpha^2+....+\alpha^{p-1}=0$ or $1+\alpha+\alpha^2+....+\alpha^{q-1}=0$ but not both together. **[IIT-JEE 2002]**

97. If $1, a_1, a_2, ..., a_{n-1}$ are the n roots of unity, then show that $(1-a_1)(1-a_2)(1-a_3)...(1-a_{n-1})=n$ **[IIT-JEE 1984]**

98. It is given that n is an odd integer greater than 3, but n is not a multiple of 3. Prove that x^3+x+x is a factor of $(x+1)^n-x^n-1$ **[IIT-JEE 1980]**

99. If $x=a+b$, $y=a\alpha+b\beta$, $z=a\beta+b\alpha$, where α,β are complex cube roots of unity, then show that $xyz=a^3+b^3$. **[IIT-JEE 1979]**

ANSWER KEY

1. (*d*) **2.** (*b*) **3.** (*d*) **4.** (*c*) **5.** [281] **6.** (*c*) **7.** (*d*) **8.** (*a*) **9.** (*b*) **10.** (*a*)

11. (*a*) **12.** (*d*) **13.** (*b*) **14.** (*a*) **15.** (*b, c*) **16.** (*a, c, d*) **17.** (*b, d*) **18.** (*a, d*) **19.** (*a, b, c*) **20.** (*b*)

21. [(A)→(q)(r), (B)→(p), (C)→(p)(s)(t), (D)→(q)(r)(s)(t)] **22.** [0.5] **23.** [4] **24.** [5] **25.** [w^2] **26.** $(a^2+b^2)(|z_1|^2+|z_2|^2)$

27. [$2n\pi$] **28.** [True] **38.** (*d*) **39.** (*c*) **40.** [A → q, B→ p] **42.** (*c*) **43.** (*c*) **44.** (*b*) **45.** (*c*)

46. (*b*) **47.** (*b*) **48.** (*d*) **49.** (*a*) **50.** (*b, d*) **51.** (*a, b, d*) **52.** (*a, c, d*) **53.** (*a, c ,d*) **54.** (*c*) **55.** (*b*)

56. [8] **57.** [False] **58.** [True] **61.** (*d*) **62.** (*d*) **63.** (*a*) **64.** (*d*) **65.** (*c*) **66.** [$z_2=-2$, $z_3=1-i\sqrt{3}$]

67. $\left[z=1-\frac{3}{2}i \text{ or } z=3-\frac{1}{2}i\right]$ **68.** [$a=b=2-\sqrt{3}$] **76.** (*d*) **77.** (*d*) **78.** (*b*) **79.** (*c*) **80.** (*d*) **81.** (*c*)

82. (*d*) **83.** (*d*) **84.** (*b*) **85.** (*b*) **86.** (*d*) **87.** (*c, d*) **88.** (*c*) **89.** [3] **90.** [4] **91.** [512]

92. [1] **93.** $\left[\left[\frac{n(n+1)}{2}\right]^2-n\right]$ **94.** [True]

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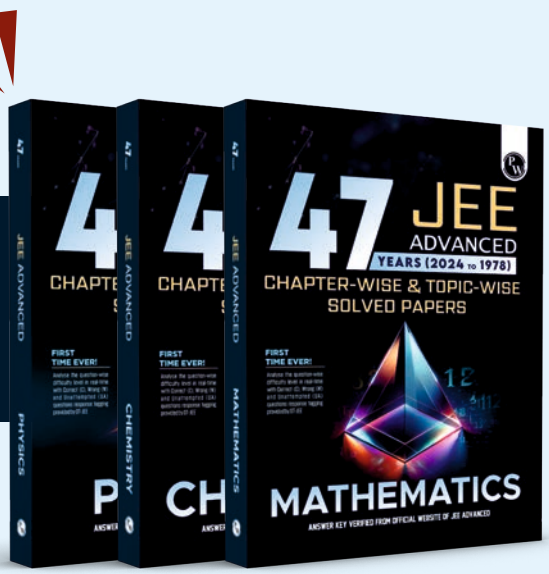

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